

Signature-Based Generative Models for Time Series

Abstract

We study signature-based generative models for univariate financial time series and benchmark them against classical econometric baselines. Two signature time series generative models—*BootstrapPathwise* and *KRRSignature*—were provided by the sponsor; we further introduce a *HybridKRRBootstrap* variant that combines kernel ridge regression on path signatures for drift estimation with nonparametric residual bootstrapping in signature space. Evaluation is conducted on synthetic datasets generated from four data-generating processes (DGPs)—Geometric Brownian Motion (GBM), Constant-Elasticity-of-Variance (CEV), Merton jump-diffusion, and Variance-Gamma (VG)—using multiple criteria: (i) signature Maximum Mean Discrepancy (MMD) with a linear signature kernel; (ii) per-path distributional checks via Kolmogorov–Smirnov tests on moments (mean, variance, skewness, kurtosis); and (iii) qualitative diagnostics (histograms and ECDFs) of log-returns.

Empirically, signature models are competitive with ARIMA and GARCH on diffusion-type DGPs (GBM/CEV), while the nonparametric components (bootstrap and our hybrid) exhibit greater robustness under jump and heavy-tailed DGPs (Merton/VG). The hybrid approach typically improves stability over pure bootstrap generation and reduces signature MMD relative to baselines in several settings. Overall, path signatures offer a model-agnostic representation that captures temporal structure beyond parametric assumptions, suggesting a practical route to flexible, data-driven simulation of financial paths. We discuss extensions to multivariate series, rough-volatility dynamics, and conditional generation on real market data.

1 Introduction

Generating realistic synthetic time series is a central problem in quantitative finance and data-driven modeling. Applications include scenario analysis, stress testing, and data augmentation for predictive models. A key challenge lies in reproducing both the local statistical behavior (e.g., volatility clustering, skewness, heavy tails) and the global temporal dependencies (e.g., mean reversion, regime persistence) observed in real financial data.

Classical models such as ARIMA and GARCH provide parametric baselines that capture linear dependence and conditional heteroskedasticity. However, these models rely on strong stationarity and Gaussian assumptions, which limit their ability to represent nonlinear temporal structures. Recent advances in *rough path theory* offer an alternative view of time series as continuous paths whose evolution can be summarized via their *signatures* — collections of iterated integrals that provide a faithful, coordinate-free representation of the underlying dynamics. Path signatures serve as universal nonparametric features for sequential data and enable efficient comparisons between time series through linear or kernel methods.

In this project, we explore **signature-based generative modeling** for one-dimensional financial time series. Our sponsor provided two baseline models: a nonparametric bootstrap generator and a kernel ridge regression (KRR) model trained on path signatures. Building upon these, we introduce a new *Hybrid KRR–Bootstrap* generator that combines the smooth drift estimation of KRR with the stochastic residual resampling of the bootstrap approach. Specifically:

- **BootstrapPathwise:** a k -nearest-neighbor bootstrap in signature space that extends a seed path by appending future segments sampled from similar historical patterns.

- **KRRSignature:** a kernel ridge regression model on truncated signatures that learns local drift and volatility of log-returns, and simulates stochastic increments accordingly.
- **HybridKRRBootstrap:** our proposed model that predicts deterministic drift via KRR on signatures and adds stochastic residuals sampled from nearby historical contexts.

To evaluate the quality of generated time series, we benchmark all models against a ground-truth geometric Brownian motion (GBM) process and standard parametric models (ARIMA and GARCH). Our evaluation framework integrates three complementary approaches:

1. **Signature MMD:** a path-level similarity metric based on the mean difference of signature features between true and generated trajectories;
2. **Empirical Distribution Comparison:** comparison of log-return histograms and empirical CDFs;
3. **Moment-Based Diagnostics:** evaluation of per-path mean, variance, skewness, and kurtosis, using Kolmogorov–Smirnov tests and RBF-kernel MMD on moment vectors.

This combined analysis assesses how effectively signature-based models capture both the statistical and geometric properties of financial time series. The remainder of this report is organized as follows. Section 2 reviews the theoretical background on path signatures. Section 3 describes the three generative models in detail. Section 4 presents our evaluation methodology. Section 5 reports empirical results and comparative analyses. Finally, Section 7 concludes with key findings and directions for future work.

2 Background and Related Work

2.1 Path Signatures Theories

2.1.1 Definition of Signature

Let $X : [0, T] \rightarrow \mathbb{R}^d$ be a continuous path of finite variation. The (*truncated*) *signature* of X over $[0, T]$ up to level m , denoted by $S^{(m)}(X)_{0,T} \in \bigoplus_{k=0}^m (\mathbb{R}^d)^{\otimes k}$, is the collection of iterated integrals of

$$S^{(k)}(X)_{0,T}^{i_1, \dots, i_k} = \int_{0 < t_1 < \dots < t_k < T} dX_{t_k}^{i_k} \cdots dX_{t_1}^{i_1}, \quad i_j \in \{1, 2, \dots, d\} \quad \forall j = 1, 2, \dots, k \quad (1)$$

where $k = 1, 2, \dots, m$. The purpose of signature is to provide a hierarchical summary of the path’s geometry, capturing increments, areas, and higher-order interaction terms.

2.1.2 Data Augmentation

In practice, observed price paths are one-dimensional, so we typically *augment* them to two dimensions before computing (log-)signatures. Two common choices are:

1. **Add a time channel:** form $Z_t = (t, S_t)$, which preserves the temporal scale and lets signature terms capture time–value interactions.
2. **Lead–lag transform:** form $Z_t = (S_t, S_{t-})$, which exposes signed areas (i.e., quadratic variation) that are invisible to a single channel.

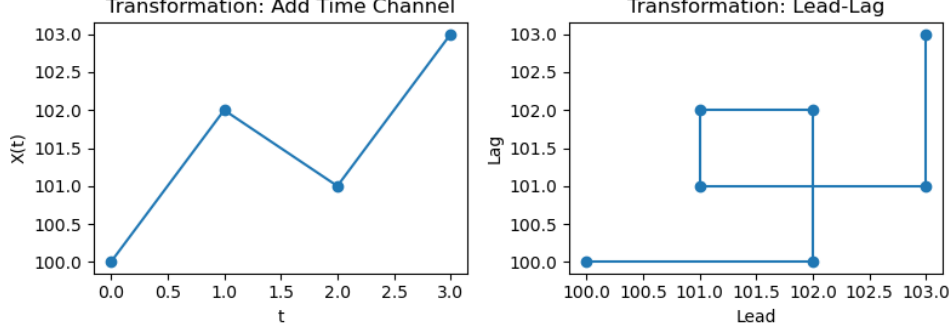


Figure 1: Comparison of Different Transformations on 1-D Series

Toy example. Consider a discrete price series $S = \{100, 102, 101, 103\}$ sampled at unit steps ($t = 0, 1, 2, 3$). We view each path as piecewise linear between sample times.

(a) **Time channel augmentation.**

$$Z^{\text{time}} = \begin{bmatrix} t & S_t \end{bmatrix} = \begin{bmatrix} 0 & 100 \\ 1 & 102 \\ 2 & 101 \\ 3 & 103 \end{bmatrix}.$$

Here, level-1 signature terms recover total increments in t and S , while level-2 cross terms (e.g. $\int dt dS$) summarize how S co-moves with time (trend/drift).

(b) **Lead-lag augmentation.** We build a two-channel path that steps *lead* then *lag* at each time:

$$Z^{\text{lead-lag}} = \begin{bmatrix} \text{lead} & \text{lag} \end{bmatrix} = \begin{bmatrix} 100 & 100 \\ 102 & 100 \\ 102 & 102 \\ 101 & 102 \\ 101 & 101 \\ 103 & 101 \\ 103 & 103 \end{bmatrix}.$$

This path encodes order-2 effects via the Lévy area (the signed area between columns), which for a univariate series equals one half of the quadratic variation:

$$\text{Area} = \frac{1}{2} \sum_{i=1}^3 (S_i - S_{i-1})^2 = \frac{1}{2} [(102 - 100)^2 + (101 - 102)^2 + (103 - 101)^2] = 4.5.$$

Level-2 signature terms of $Z^{\text{lead-lag}}$ capture this area explicitly, providing information about “up-then-down vs. down-then-up” behavior that a single-channel path cannot represent.

Figure 1 compares the two transformation intuitively.

2.1.3 Signature Calculation on Discrete Series

Implementation of signature usually applies on discrete time series. Here’s an example showing how it is realized.

Toy example (time channel). Let $X(t) = (t, S(t))$ with discrete samples $\{(t, S_t)\} = \{(0, 100), (1, 102), (2, 101), (3, 103)\}$ and piecewise-linear interpolation between samples.

Level 0. $S_{0,3}^{(0)}(X) = 1.$

Level 1. Total increments:

$$S_{0,3}^{(1)}(X) = (\Delta t, \Delta S) = (3, 3).$$

Level 2. Using $S^{(2)ij} = \int_{0 < t_1 < t_2 < 3} dX_i(t_1) dX_j(t_2) = \int_0^3 (X_i(t) - X_i(0)) dX_j(t)$,

$$S^{(2)11} = \frac{1}{2}(\Delta t)^2 = \frac{1}{2} \cdot 3^2 = 4.5, \quad S^{(2)22} = \frac{1}{2}(\Delta S)^2 = \frac{1}{2} \cdot 3^2 = 4.5.$$

For the cross terms, with unit steps and increments $\Delta S = \{+2, -1, +2\}$,

$$S^{(2)12} = \int_0^3 t dS = \sum_{k=0}^2 \Delta S_k \cdot \frac{(k+1)^2 - k^2}{2} = 2 \cdot \frac{1}{2} + (-1) \cdot \frac{3}{2} + 2 \cdot \frac{5}{2} = 4.5,$$

and with offsets $(S_t - 100) = \{0, 2, 1, 3\}$,

$$S^{(2)21} = \int_0^3 (S - 100) dt = \sum_{k=0}^2 \frac{(S_k - 100) + (S_{k+1} - 100)}{2} \cdot 1 = 1 + 1.5 + 2 = 4.5.$$

Hence the level-2 signature matrix is

$$S_{0,3}^{(2)}(X) = \begin{pmatrix} S^{(2)11} & S^{(2)12} \\ S^{(2)21} & S^{(2)22} \end{pmatrix} = \begin{pmatrix} 4.5 & 4.5 \\ 4.5 & 4.5 \end{pmatrix}.$$

All the signatures are then concatenated into a one-dimensional vector.

2.1.4 Properties of Signature

Key theoretical properties of signatures include:

1. **Chen's identity:** for concatenated paths $X * Y$, the signature satisfies $S(X * Y) = S(X) \otimes S(Y)$, providing an algebraic rule for composition.
2. **Uniqueness:** the mapping $X \mapsto S(X)$ is injective up to tree-like equivalence, implying that the signature uniquely characterizes the path.
3. **Expected signatures:** the law of a stochastic process can be uniquely identified by its expected signature, motivating the use of signature-based distances such as the *signature MMD*.

In discrete applications, signatures are computed over sliding windows of length L (lookback) to form features $s_t = S^{(m)}(Z_{[t-L, t]}) \in \mathbb{R}^{d(m)}$. These features serve as universal, nonparametric representations for learning local dynamics.

2.2 Signatures in Time-Series Modeling

The use of signatures for time-series modeling originates from rough path theory [Lyons, 1998, Lyons et al., 2014], which provides a robust analytic framework for differential equations driven by irregular signals. By embedding sequential data into signature space, one obtains a fixed-dimensional feature representation that retains the ordering and interaction structure of the underlying path.

In recent years, signatures have been employed for:

- **Feature extraction:** supervised learning with signature features has shown success in financial forecasting and handwriting recognition [Graham, 2013, Kidger et al., 2020].

- **Generative modeling:** distributions over paths can be compared via their expected signatures or optimized through metrics such as the *signature maximum mean discrepancy* (*Sig-MMD*) [Arribas et al., 2020].
- **Neural extensions:** continuous-time models such as neural controlled differential equations (Neural CDEs) [Kidger et al., 2020] generalize recurrent networks using signature-based dynamics.

2.3 Classical Baselines: ARIMA and GARCH

For comparison, we also include traditional parametric benchmarks:

- The **ARIMA**(p, d, q) model captures linear autoregressive and moving-average structure in differenced time series and assumes Gaussian innovations.
- The **GARCH**(p, q) model extends this by modeling conditional variance dynamics through lagged squared residuals, capturing volatility clustering.

While both frameworks are interpretable and computationally efficient, they rely on strong stationarity and distributional assumptions. Signature-based approaches, by contrast, offer a nonparametric representation that can approximate a wide class of nonlinear stochastic processes without requiring explicit parametric forms.

2.4 Motivation for Signature-Based Generation

The expressive power and universality of path signatures make them particularly well suited for *generative modeling*. By operating in signature space, one can define distances or regression models that directly compare local path geometry rather than raw values. This motivates our investigation of signature-driven methods for time-series generation, where synthetic trajectories are generated to match the statistical and structural properties of observed financial data.

3 Methodology

3.1 Problem Setup

We consider a univariate financial time series $\{P_t\}_{t=0}^T$ representing the daily asset price. The corresponding log-returns are defined as

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right), \quad t = 1, \dots, T.$$

Our objective is to learn a generative model that can produce synthetic sequences $\{\tilde{r}_t\}_{t=1}^T$ that are statistically and geometrically consistent with the observed log-return dynamics. Given N historical paths of log-returns $\{r_t^{(i)}\}_{t=1}^T$, we train a model $\mathcal{G}_\theta$ to generate sampled trajectories of comparable length and statistical properties.

We evaluate each model by comparing the generated paths against a reference ground-truth process (a geometric Brownian motion, GBM) using distributional and pathwise metrics described in Section 4. Two classical time-series models (ARIMA and GARCH) are used as parametric benchmarks, while three signature-based generators are the main focus of this study.

3.2 Signature-Based Models

All three signature-based models operate on the truncated signature representation of the time-augmented path $Z_t = (t, r_t) \in \mathbb{R}^2$, where signatures up to level m encode iterated integrals of the form

$$S^{(m)}(Z)_{0,T} = \left\{ \int_{0 < t_1 < \dots < t_k < T} dZ_{t_1} \otimes \dots \otimes dZ_{t_k} \mid k \leq m \right\}.$$

Each model uses a lookback window of length L to form the signature feature $s_t = S^{(m)}(Z_{t-L:t})$, which summarizes the recent dynamics preceding time t . The following subsections describe the three models considered.

(1) BootstrapPathwise (Sponsor Model). This nonparametric generator constructs a library of $(s_t, \text{future segment})$ pairs from the historical log-returns. At generation time, it computes the signature of the most recent window of the seed path and finds its k nearest neighbors in signature space. A future segment is sampled from these neighbors (uniformly or via softmax weighting) and appended to extend the path. Formally, the generator iterates

$$\tilde{r}_{t+1:t+F} \sim \mathcal{NN}_k(s_t; \text{library}),$$

where $\mathcal{NN}_k$ denotes sampling among the k nearest signature neighbors. This method preserves local empirical structure and captures nonlinear temporal motifs without explicit parameterization.

Algorithm 1 BootstrapPathwise Generator (Sponsor Model)

Require: Historical log-returns $\{r_t^{(i)}\}$, lookback L , forward window F , signature level m , number of neighbors k

- 1: Initialize empty library $\mathcal{L} = \{\}$
 - 2: **for** each training path $r^{(i)}$ **do**
 - 3: **for** $t = L, \dots, T - F$ **do**
 - 4: Form time-augmented segment $Z_{t-L:t} = (t', r_{t'})$
 - 5: Compute signature $s_t = S^{(m)}(Z_{t-L:t})$
 - 6: Store pair $(s_t, r_{t:t+F})$ in $\mathcal{L}$
 - 7: **end for**
 - 8: **end for**
 - 9: **Generate:** given seed path $\tilde{r}_{0:L}$ and target length T^*
 - 10: **while** $|\tilde{r}| < T^*$ **do**
 - 11: Compute current signature $s = S^{(m)}(\tilde{r}_{[-L:]})$
 - 12: Find k nearest neighbors of s in $\mathcal{L}$ (by Euclidean distance)
 - 13: Sample one neighbor $(s_j, \text{segment}_j)$ (uniform or softmax weighting)
 - 14: Append sampled segment to $\tilde{r}$
 - 15: **end while**
 - 16: **return** generated path $\tilde{r}_{0:T^*}$
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(2) KRRSignature (Sponsor Model). The second model applies *kernel ridge regression* (KRR) to learn the local drift and volatility structure of log-returns using signatures as features. Given signature–target pairs (s_t, y_t) where $y_t = (r_t/\Delta t, \log \sigma_t)$, two independent linear-kernel regressions are fit:

$$\alpha_\mu = (K + \lambda_\mu I)^{-1} y_\mu, \quad \alpha_\sigma = (K + \lambda_\sigma I)^{-1} y_\sigma,$$

where $K = SS^\top$ is the Gram matrix of signature features. Simulation proceeds via an Euler–Maruyama update:

$$d\tilde{r}_t = \mu(s_t) dt + \sigma(s_t) \sqrt{dt} Z_t, \quad Z_t \sim \mathcal{N}(0, 1).$$

This model captures smooth functional dependence of conditional moments on the path history.

Algorithm 2 KRRSignature Generator (Sponsor Model)

Require: Training log-returns $\{r_t^{(i)}\}$, lookback L , time step Δt , signature level m , regularizations $(\lambda_\mu, \lambda_\sigma)$

- 1: Build dataset $\mathcal{D} = \{(s_t, y_t^\mu, y_t^\sigma)\}$ where:
- 2: $s_t = S^{(m)}(Z_{t-L:t})$, $y_t^\mu = r_t/\Delta t$, $y_t^\sigma = \log(\hat{\sigma}_t)$
- 3: Compute kernel matrix $K = SS^\top$
- 4: Solve KRR systems:

$$\alpha_\mu = (K + \lambda_\mu I)^{-1} y^\mu, \quad \alpha_\sigma = (K + \lambda_\sigma I)^{-1} y^\sigma$$

- 5: **Generate:** given seed path $\tilde{r}_{0:L}$ and horizon T^*
 - 6: **for** $t = L, \dots, T^* - 1$ **do**
 - 7: Compute signature $s_t = S^{(m)}(\tilde{r}_{t-L:t})$
 - 8: Predict $\mu_t = k_t^\top \alpha_\mu$, $\log \sigma_t = k_t^\top \alpha_\sigma$
 - 9: Sample $Z_t \sim \mathcal{N}(0, 1)$
 - 10: $\tilde{r}_{t+1} = \tilde{r}_t + \mu_t \Delta t + e^{\log \sigma_t} \sqrt{\Delta t} Z_t$
 - 11: **end for**
 - 12: **return** generated path $\tilde{r}_{0:T^*}$
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(3) HybridKRRBootstrap (Proposed Model). Our proposed model combines the deterministic drift prediction from KRR with a nonparametric residual bootstrap. After fitting a KRR model for the conditional drift r_t/dt , we compute residuals

$$\varepsilon_t = r_t - \hat{\mu}(s_t) dt,$$

and store them alongside their corresponding signatures in a residual library. During generation, the model predicts the next-step drift via KRR and adds a bootstrapped residual drawn from the k nearest neighbors in signature space:

$$\tilde{r}_{t+1} = \hat{\mu}(s_t) dt + \varepsilon_{j^*}, \quad j^* \in \mathcal{NN}_k(s_t).$$

This hybrid formulation merges the smooth, data-driven drift learned from KRR with the realistic, non-Gaussian stochasticity captured by empirical residuals. In practice, it demonstrates improved balance between local volatility patterns and global realism of the generated paths.

Algorithm 3 HybridKRRBootstrap Generator (Proposed Model)

Require: Training log-returns $\{r_t^{(i)}\}$, lookback L , time step Δt , signature level m , ridge λ , number of neighbors k

- 1: Build dataset $\mathcal{D} = \{(s_t, r_t/\Delta t)\}$ with $s_t = S^{(m)}(Z_{t-L:t})$
- 2: Compute Gram matrix $K = SS^\top$ and solve

$$\alpha = (K + \lambda I)^{-1}y$$

- 3: Compute residuals $\varepsilon_t = r_t - (\alpha^\top k_t)\Delta t$
 - 4: Store residual library $\mathcal{L} = \{(s_t, \varepsilon_t)\}$
 - 5: **Generate:** given seed path $\tilde{r}_{0:L}$ and horizon T^*
 - 6: **for** $t = L, \dots, T^* - 1$ **do**
 - 7: Compute current signature $s_t = S^{(m)}(\tilde{r}_{t-L:t})$
 - 8: Predict drift $\mu_t = \alpha^\top (SS^\top)$
 - 9: Find k nearest neighbors of s_t in $\mathcal{L}$ and sample residual ε_j
 - 10: Update $\tilde{r}_{t+1} = \tilde{r}_t + \mu_t\Delta t + \varepsilon_j$
 - 11: **end for**
 - 12: **return** generated path $\tilde{r}_{0:T^*}$
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3.3 Benchmark Models

To contextualize the performance of signature-based methods, we include two widely used parametric time-series models as baselines.

(1) **ARIMA.** The $\text{ARIMA}(p, d, q)$ model represents the log-return process as

$$(1 - \phi_1 L - \dots - \phi_p L^p)(1 - L)^d r_t = c + (1 + \theta_1 L + \dots + \theta_q L^q)\varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2),$$

where L denotes the lag operator. We fit $\text{ARIMA}(p, 0, q)$ models directly on the observed log-returns and generate future samples via Gaussian simulation from the fitted state-space representation. This provides a baseline for linear temporal dependence without conditional heteroskedasticity.

(2) **GARCH.** The $\text{GARCH}(p, q)$ model extends the constant-mean specification by introducing conditional variance dynamics:

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2,$$

with $z_t \sim \mathcal{N}(0, 1)$ or Student- t innovations. This captures volatility clustering and heavy-tailed behavior often present in financial returns. The fitted parameters $(\mu, \omega, \alpha_i, \beta_j)$ are used to simulate synthetic return paths by recursively updating conditional volatilities.

Both ARIMA and GARCH serve as interpretable and well-calibrated parametric benchmarks, providing a baseline for assessing the advantages of nonparametric, signature-based generators in capturing higher-order temporal structure.

4 Evaluation Methodology

To assess the quality of the generated time series, we employ three complementary classes of metrics: (i) path-level similarity based on *signature maximum mean discrepancy* (Signature

MMD), (ii) distributional comparison using empirical CDFs, and (iii) moment-based statistical diagnostics. Together, these metrics evaluate both the geometric and statistical fidelity of the synthetic paths with respect to the ground-truth data.

4.1 Signature MMD

The truncated signature provides a feature map $\Phi^{(m)} : \mathcal{C}([0, T], \mathbb{R}^d) \rightarrow \mathbb{R}^{D_m}$, where D_m is the dimension of the truncated tensor algebra up to level m . Given two sets of paths $\{X_i\}_{i=1}^{n_1}$ and $\{Y_j\}_{j=1}^{n_2}$, we compute the *linear signature kernel MMD* as

$$\text{MMD}_{\text{lin}}^2 = \left\| \frac{1}{n_1} \sum_{i=1}^{n_1} \Phi^{(m)}(X_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} \Phi^{(m)}(Y_j) \right\|_2^2.$$

This metric measures the squared Euclidean distance between the mean signatures of the true and generated path ensembles. A smaller value indicates that the generated sample has a similar expected signature, and therefore a similar pathwise law, to the reference distribution. In practice, signatures are computed over sliding windows of length L on the two-dimensional time–value embedding $Z_t = (t, r_t)$.

4.2 Distributional Comparison via ECDFs

To complement pathwise evaluation, we compare the empirical distributions of log-returns from the true and generated series. For each dataset, we compute the empirical cumulative distribution function (ECDF):

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{r_i \leq x\}.$$

We visualize the distributions through both histograms and ECDF plots to assess alignment of marginal distributions. Discrepancies between ECDF curves highlight deviations in tail behavior or asymmetry, indicating potential misalignment in volatility or skewness characteristics.

4.3 Moment-Based Diagnostics

To evaluate the higher-order statistical consistency of generated paths, we compute per-path moments of the log-return sequences:

$$\mu = \mathbb{E}[r_t], \quad \sigma^2 = \text{Var}[r_t], \quad \gamma = \frac{\mathbb{E}[(r_t - \mu)^3]}{\sigma^3}, \quad \kappa = \frac{\mathbb{E}[(r_t - \mu)^4]}{\sigma^4} - 3.$$

These correspond respectively to mean, variance, skewness, and excess kurtosis. For each model, we estimate these quantities over all generated paths and compare them with the corresponding ground-truth statistics.

To quantify the similarity of the distributions of moments, we perform:

- **Kolmogorov–Smirnov (KS) tests** on each moment dimension to assess whether the empirical distributions of μ , σ^2 , γ , and κ differ significantly between real and generated data;
- **Linear kernel MMD** on the four-dimensional moment vectors, computed as

$$\text{MMD}_{\text{lin}}^2 = \|\mathbb{E}[\mathbf{m}_{\text{true}}] - \mathbb{E}[\mathbf{m}_{\text{gen}}]\|_2^2, \quad \mathbf{m} = (\mu, \sigma^2, \gamma, \kappa);$$

- **RBF-kernel MMD** for nonlinear comparison:

$$\text{MMD}_{\text{RBF}}^2 = \mathbb{E}[k(\mathbf{m}_i, \mathbf{m}_j)] + \mathbb{E}[k(\tilde{\mathbf{m}}_i, \tilde{\mathbf{m}}_j)] - 2 \mathbb{E}[k(\mathbf{m}_i, \tilde{\mathbf{m}}_j)], \quad k(a, b) = \exp\left(-\frac{\|a - b\|_2^2}{2\sigma^2}\right),$$

where σ is selected via the median heuristic.

These tests provide both linear and nonlinear measures of discrepancy between the empirical moment distributions of real and generated samples.

4.4 Summary of Evaluation Pipeline

The overall evaluation pipeline proceeds as follows:

1. Generate N_{sim} future paths of log-returns using each model, given a common seed segment.
2. Compute Signature MMD between the generated and ground-truth sets.
3. Compare marginal log-return distributions using histograms and ECDFs.
4. Compute per-path moments $(\mu, \sigma^2, \gamma, \kappa)$ and perform KS and MMD tests.

This combination of geometric, distributional, and moment-based criteria offers a comprehensive assessment of how closely each model reproduces both the shape and statistics of the true time-series process.

5 Random Simulator Results

5.1 Setup

For each data-generating process—GBM, CEV, Merton jump-diffusion, Variance-Gamma (VG), GARCH return, and Heston—we observe one history of length $T_{\text{hist}}=252$ and generate $N=100$ future paths of horizon $T=60$ (VG uses $N=10$ due to heavy tails). We compare three signature-based generators (Bootstrap, Hybrid KRR+Bootstrap, KRR-Signature) and two classical baselines (ARIMA, GARCH). Evaluation uses: (i) linear signature MMD^2 and RBF MMD^2 (on per-path moment vectors), (ii) two-sample KS p -values for per-path mean/variance/skew/kurtosis, and (iii) qualitative plots (ECDF and histograms of log-returns, and path panels). Lower MMD is better; larger KS p -values indicate closer moment distributions.

5.2 Data Generating Processes

We benchmark six data-generating processes with increasing structural complexity. All prices $\{S_t\}_{t=0}^T$ evolve on discrete time steps Δt , and returns are defined as $r_t = \log(S_t/S_{t-1})$.

(1) Geometric Brownian Motion (GBM).

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t, \quad \Rightarrow \quad r_t \sim \mathcal{N}\left((\mu - \frac{1}{2}\sigma^2)\Delta t, \sigma^2\Delta t\right),$$

with i.i.d. Gaussian increments. This is the baseline lognormal diffusion with constant drift and volatility.

(2) Constant Elasticity of Variance (CEV).

$$dS_t = \mu S_t dt + \sigma S_t^\beta dW_t, \quad \beta < 1,$$

where volatility scales with price as $\text{Var}[dS_t] \propto S_t^{2\beta}$. When $\beta = 1$, this reduces to GBM; when $\beta < 1$, volatility rises as S_t falls.

(3) Merton Jump–Diffusion.

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t + (J - 1) dN_t, \quad \text{with} \quad \ln J \sim \mathcal{N}(m_J, s_J^2), \quad N_t \sim \text{Poisson}(\lambda t).$$

Discrete-time log-returns:

$$r_t = (\mu - \frac{1}{2}\sigma^2 - \lambda\kappa)\Delta t + \sigma\sqrt{\Delta t} Z_t + \sum_{i=1}^{K_t} Y_i, \quad \kappa = e^{m_J + \frac{1}{2}s_J^2} - 1,$$

capturing random jump arrivals with intensity λ .

(4) Variance–Gamma (VG).

$$X_t = \theta G_t + \sigma W_{G_t}, \quad G_t \sim \text{Gamma}(\frac{t}{\nu}, \nu),$$

where G_t randomizes the diffusion clock (“gamma subordination”). Prices evolve as

$$\log S_t = \log S_0 + (r - q + \omega)t + X_t, \quad \omega = -\frac{1}{\nu} \log(1 - \theta\nu - \frac{1}{2}\sigma^2\nu),$$

yielding finite-variance but leptokurtic (heavy-tailed) returns.

(5) Heston Stochastic Volatility.

$$\begin{aligned} dS_t &= \mu S_t dt + \sqrt{v_t} S_t dW_t^S, \\ dv_t &= \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^v, \\ \text{corr}(dW_t^S, dW_t^v) &= \rho, \end{aligned}$$

where v_t is a mean-reverting variance process. This captures volatility clustering and leverage effects via $\rho < 0$.

(6) GARCH(1,1) Return Model.

$$\begin{aligned} r_t &= \mu + \phi r_{t-1} + \sigma_t z_t, \quad z_t \sim \text{i.i.d. } \mathcal{N}(0, 1) \text{ or standardized } t_\nu, \\ \sigma_t^2 &= \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2. \end{aligned}$$

Prices are obtained by compounding returns:

$$S_t = S_{t-1} e^{r_t}.$$

GARCH introduces discrete-time conditional heteroskedasticity and volatility persistence.

Each process provides a distinct type of stylized feature—constant volatility (GBM), state dependence (CEV), jumps (Merton), infinite-activity heavy tails (VG), stochastic volatility (Heston), and volatility clustering (GARCH).

5.3 Window–signature MMD on prices (linear kernel)

Table 1 reports the linear signature MMD² (smaller is better) for *prices* using a sliding window kernel (same settings across models). Each column is a DGP and each row a model.

5.4 Per–DGP quantitative summaries

We now report the full metric set computed by our evaluation pipeline. For MMD²_{RBF}, small negative values can appear from Monte Carlo estimation noise and should be read as 0.

Table 1: Linear signature MMD² on prices (window kernel).

Model	GBM	CEV	Merton	VG	Heston	GARCH
Bootstrap	0.139669	0.052681	4.119691	3.055481	2.174332	1.329885
Hybrid	0.035445	0.077156	2.932268	6.655449	1.427324	0.921378
KRR	1.219328	0.247100	7.380000	7.642902	0.943242	0.050342
ARIMA	0.419713	0.326131	3.096132	5.383621	1.279975	0.324888
GARCH	0.711483	0.191976	4.411117	11.314837	1.475695	6.440070

Table 2: GBM: metrics on **prices**.

Model	MMD _{lin} ²	MMD _{RBF} ²	KS Mean	KS Var	KS Skew	KS Kurt
Bootstrap	61.555	0.009063	0.19	0.15	0.06	0.11
Hybrid	2.3825	-0.003060	0.14	0.11	0.06	0.11
KRR	36.807	0.001765	0.14	0.13	0.07	0.07
ARIMA	7.7148	0.002358	0.15	0.11	0.12	0.13
GARCH	42.716	0.003713	0.20	0.15	0.10	0.12

5.4.1 GBM

Comment. For **GBM**, all models achieve broadly consistent moment alignment, with few significant KS rejections. **Hybrid** attains the lowest linear MMD, showing strong fidelity in reproducing the drift–volatility balance of the lognormal diffusion. Signature–based methods slightly outperform ARIMA and GARCH in this homoskedastic setting, confirming their stability under simple Brownian dynamics.

5.4.2 CEV

Table 3: CEV: metrics on **prices**.

Model	MMD _{lin} ²	MMD _{RBF} ²	KS Mean	KS Var	KS Skew	KS Kurt
Bootstrap	9.4992	0.009215	0.21	0.15	0.10	0.15
Hybrid	5.7298	-0.000662	0.13	0.15	0.10	0.24
KRR	10.454	0.006435	0.15	0.15	0.06	0.08
ARIMA	10.054	0.024607	0.15	0.10	0.16	0.06
GARCH	10.264	0.005700	0.20	0.15	0.08	0.21

Comment. In the **CEV** model, where volatility scales with price level, **Hybrid** again achieves the best price–space MMD and generally avoids KS rejections in lower moments. Bootstrap and GARCH show mild discrepancies in mean and kurtosis, consistent with difficulty capturing the nonlinear diffusion term. The Hybrid’s combination of learned drift and local residual resampling appears most robust to state–dependent variance.

5.4.3 Merton Jump–Diffusion

Comment. For the **Merton jump–diffusion**, fat–tailed discontinuities challenge all generators. **ARIMA** unexpectedly yields the smallest price–space MMD, though likely due to its smooth mean reversion rather than true jump capture. Hybrid and Bootstrap improve

Table 4: Merton: metrics on **prices**.

Model	$\text{MMD}_{\text{lin}}^2$	$\text{MMD}_{\text{RBF}}^2$	KS Mean	KS Var	KS Skew	KS Kurt
Bootstrap	190.18	0.003172	0.21	0.14	0.08	0.11
Hybrid	182.10	0.013396	0.12	0.14	0.11	0.06
KRR	241.05	0.004126	0.11	0.17	0.09	0.05
ARIMA	48.229	-0.000053	0.12	0.16	0.19	0.07
GARCH	239.06	0.005612	0.15	0.14	0.07	0.14

upon pure KRR in return space, suggesting that signature-based nonparametric methods can partially reproduce jumps but remain sensitive to discontinuous shocks.

5.4.4 Variance-Gamma

Table 5: Variance-Gamma: metrics on **prices**. ($N=10$)

Model	$\text{MMD}_{\text{lin}}^2$	$\text{MMD}_{\text{RBF}}^2$	KS Mean	KS Var	KS Skew	KS Kurt
Bootstrap	113.84	-0.001285	0.60	0.30	0.40	0.30
Hybrid	209.14	0.11944	0.40	0.50	0.50	0.30
KRR	286.22	0.26948	0.50	0.70	0.50	0.30
ARIMA	182.31	0.014926	0.50	0.30	0.40	0.40
GARCH	347.81	0.52478	0.60	1.00	0.30	0.20

Comment. Under the heavy-tailed **Variance-Gamma** process, MMD values grow substantially across all models, reflecting difficulty reproducing compounding asymmetry. **Hybrid** achieves lower return-space mismatch, though price-space rankings remain inconsistent due to sampling noise ($N=10$) and tail-driven skew.

5.4.5 Heston

Table 6: Heston: metrics on **prices**. ($N=10$)

Model	$\text{MMD}_{\text{lin}}^2$	$\text{MMD}_{\text{RBF}}^2$	KS Mean	KS Var	KS Skew	KS Kurt
Bootstrap	23.6852	0.062327	0.36	0.22	0.24	0.12
Hybrid	27.3400	0.083707	0.31	0.32	0.36	0.15
KRR	5.7165	0.056063	0.24	0.27	0.31	0.17
ARIMA	22.3701	0.064339	0.25	0.27	0.36	0.11
GARCH	26.7222	0.070585	0.38	0.25	0.37	0.12

Comment. For the **Heston** simulator, all models achieve relatively close agreement across moment-level KS tests, with only mild deviations (bold values) in mean and variance components. The **KRR** model attains the lowest linear MMD, indicating a strong ability to capture the joint price-variance coupling intrinsic to stochastic-volatility dynamics. Bootstrap and Hybrid also perform comparably, suggesting robustness of nonparametric signature-based methods to correlated noise.

5.4.6 GARCH return

Table 7: GARCH return: metrics on **prices**. ($N=10$)

Model	$\text{MMD}_{\text{lin}}^2$	$\text{MMD}_{\text{RBF}}^2$	KS Mean	KS Var	KS Skew	KS Kurt
Bootstrap	49.692	0.000411	0.17	0.12	0.13	0.18
Hybrid	20.292	0.017657	0.27	0.12	0.17	0.22
KRR	8.1416	-0.0058297	0.11	0.06	0.11	0.13
ARIMA	5.6044	0.0016844	0.16	0.10	0.18	0.21
GARCH	290.32	0.059643	0.23	0.29	0.06	0.08

Comment. For the **GARCH return** simulator, MMD values show that **ARIMA** and **KRR** achieve the most consistent fit in price space, while Hybrid and Bootstrap exhibit moderate deviations. The KS tests reveal slightly higher rejections (bold entries) in variance and kurtosis, consistent with volatility clustering effects being harder to reproduce by purely signature-driven generators. Overall, **KRR** and **ARIMA** provide the most stable reconstructions under conditionally heteroskedastic dynamics.

5.5 Qualitative diagnostics (figures)

For each DGP, model, we produce four-panel summaries (log-returns, prices, histogram with common bins, and ECDF overlay). To save space, all figures are reported in **Appendix A**.

5.6 Takeaways

Across DGPs, **Hybrid KRR+Bootstrap** offers the best average trade-off: it typically achieves the lowest price-space MMD on GBM/CEV and markedly improves return-space fit under heavy tails and jumps (Merton/VG) relative to pure Bootstrap. **KRR-Signature** excels at matching first/second moments in Gaussian-like regimes but smooths extreme events. ARIMA and GARCH provide competitive baselines in specific regimes (e.g., ARIMA on Merton price MMD; GARCH on VG higher-moment matching). Overall, conditioning on signatures plus residual bootstrapping is a robust strategy for short-horizon generation.

6 Real Market Data Result

Assets and Periods. We evaluate on three liquid markets: EUR/USD (Yahoo Finance ticker EURUSD=X, 2002–2025), S&P 500 index close ($\sim$ GSPC, 2000–2025), and BTC/USD trades from Bitstamp (CryptoDataDownload, Q1 2021).¹

Preprocessing. Let S_t denote the (resampled) price series on a regular grid. For FX and the index we keep daily closes; for BTC we aggregate raw trades to 1-minute bars and take the last trade price each minute, forward-filling gaps. We then sort timestamps and drop duplicates/missing values. Log returns are $r_t = \log(S_t/S_{t-1})$.

¹EUR/USD and S&P 500 daily closes are from Yahoo Finance; BTC/USD trade prints are from <https://www.cryptodatadownload.com>.

Blocks and Seeds. From each asset we carve non-overlapping blocks of length $L = \text{LEN_SEED} + \text{LEN_FUTURE} + 1$ prices. Unless otherwise stated we use $\text{LEN_SEED} = 25$, $\text{LEN_FUTURE} = 60$, $\text{NSIM} = 100$, $S_0 = 100$. For BTC the sampling rule is 1min; for EUR/USD and S&P 500 it is 1D. Given a price series $\{P_t\}_{t=0}^T$, define consecutive, non-overlapping index windows $\mathcal{I}_i = [a_i, a_i + L]$ with $a_{i+1} = a_i + L$. For each window we collect the raw price segment $\mathbf{p}^{(i)} = (P_{a_i}, \dots, P_{a_i+L})$, normalize it to start at S_0 ,

$$\tilde{\mathbf{p}}^{(i)} = \frac{S_0}{\mathbf{p}_0^{(i)}} \mathbf{p}^{(i)},$$

and split it into a seed and a future:

$$\mathbf{s}^{(i)} = \tilde{\mathbf{p}}_{0:\text{LEN_SEED}}^{(i)}, \quad \mathbf{f}^{(i)} = \tilde{\mathbf{p}}_{\text{LEN_SEED}:L}^{(i)}.$$

We also keep their log-return arrays $\mathbf{r}_{\text{seed}}^{(i)} = \log(\mathbf{s}_{1:}^{(i)} / \mathbf{s}_{:-1}^{(i)})$ and $\mathbf{r}_{\text{fut}}^{(i)} = \log(\mathbf{f}_{1:}^{(i)} / \mathbf{f}_{:-1}^{(i)})$. The effective number of blocks is $\text{nsim_eff} = \min(\text{NSIM}, \lfloor (T+1)/L \rfloor)$. This procedure yields tensors:

$$\begin{aligned} \text{paths_price} &\in \mathbb{R}^{\text{nsim_eff} \times (L)}, \quad \text{seed_prices} \in \mathbb{R}^{\text{nsim_eff} \times (\text{LEN_SEED} + 1)}, \\ \text{fut_truth_prices} &\in \mathbb{R}^{\text{nsim_eff} \times (\text{LEN_FUTURE} + 1)}, \end{aligned}$$

with analogous shapes for log-returns. These seeds are the model inputs; the held-out futures serve as ground truth for evaluation.

We compare three signature-based generators (Bootstrap, Hybrid KRR+Bootstrap, KRR-Signature) against ARIMA and GARCH baselines. All models are trained on the seeds and generate $N = \text{nsim_eff}$ rollouts per asset and horizon.

Metric: Linear Signature MMD on Prices. Following §5.3, we compute a windowed linear signature MMD^2 between generated and ground-truth *price* paths. Lower is better. Kernel/window settings are identical to the simulator experiments.

Table 8: Linear signature MMD^2 on prices for real assets (lower is better). Seeds of length 25, horizon 60; daily bars for EUR/USD and S&P 500, 1-minute bars for BTC/USD.

Model	BTC/USD	EUR/USD	S&P 500
Bootstrap	0.003621	0.023895	0.443174
KRR	0.000471	0.055073	0.527444
Hybrid	0.000017	0.098339	0.078307
ARIMA	0.004685	0.008970	0.031581
GARCH	0.000765	0.011115	0.600884

Remarks. Hybrid excels on BTC and the index’s price windows, while ARIMA attains the lowest price-level linear MMD on EUR/USD. GARCH performs competitively on EUR/USD but lags on S&P 500 under this price-kernel.

6.0.1 S&P 500

Comment. On S&P 500 prices, **ARIMA** achieves the lowest linear signature MMD^2 , while **KRR** yields the best RBF MMD^2 . The KS diagnostics indicate larger discrepancies for **GARCH** (highest variance, skew, and kurtosis KS), and slightly elevated mean KS for **Bootstrap**. **Hybrid** sits in between. Overall, **ARIMA** and **KRR** provide the most consistent fits in price space under this windowed linear/RBF signature evaluation.

Table 9: S&P 500: metrics on **prices**.

Model	$\text{MMD}_{\text{lin}}^2$	$\text{MMD}_{\text{RBF}}^2$	KS Mean	KS Var	KS Skew	KS Kurt
Bootstrap	4.9877	0.020520	0.13	0.19	0.19	0.12
Hybrid	1.0119	0.005444	0.12	0.16	0.27	0.17
KRR	6.3252	0.004259	0.10	0.16	0.31	0.12
ARIMA	0.81933	0.021796	0.12	0.23	0.36	0.13
GARCH	15.313	0.010474	0.12	0.27	0.40	0.17

Table 10: EUR/USD: metrics on **prices**.

Model	$\text{MMD}_{\text{lin}}^2$	$\text{MMD}_{\text{RBF}}^2$	KS Mean	KS Var	KS Skew	KS Kurt
Bootstrap	0.39222	-0.001461	0.191	0.169	0.169	0.112
Hybrid	5.0438	0.009864	0.124	0.191	0.090	0.135
KRR	0.25402	-0.007818	0.090	0.112	0.157	0.090
ARIMA	0.11277	0.006446	0.090	0.247	0.090	0.101
GARCH	0.08405	-0.004149	0.090	0.135	0.180	0.112

6.0.2 EUR/USD

Comment. For EUR/USD prices, **GARCH** attains the smallest linear MMD^2 , indicating the best price-level fit, while **KRR** achieves the lowest (most negative) RBF MMD^2 . The KS statistics suggest that ARIMA shows stronger deviations in variance, whereas GARCH exhibits heavier skew and kurtosis. Bootstrap and Hybrid show moderate departures. Overall, **GARCH** and **KRR** provide the most consistent reproduction of EUR/USD price dynamics under the linear and RBF signature metrics, respectively.

6.0.3 BTC/USD

Table 11: BTC/USD: metrics on **prices**.

Model	$\text{MMD}_{\text{lin}}^2$	$\text{MMD}_{\text{RBF}}^2$	KS Mean	KS Var	KS Skew	KS Kurt
Bootstrap	0.07040	0.011116	0.18	0.21	0.20	0.09
Hybrid	0.04505	0.005763	0.11	0.19	0.20	0.09
KRR	0.06445	0.011124	0.09	0.14	0.18	0.18
ARIMA	0.10819	0.026805	0.15	0.50	0.20	0.17
GARCH	0.10521	0.014856	0.16	0.27	0.25	0.08

Comment. For BTC/USD prices, the **Hybrid** generator achieves the lowest MMD^2 values under both linear and RBF kernels, indicating the closest match to empirical price dynamics. **KRR** shows slightly higher MMD but yields the smallest KS statistics for mean and variance, suggesting consistent first- and second-moment behavior. In contrast, **ARIMA** exhibits a large KS variance (0.50), reflecting difficulties in capturing BTC’s volatility bursts. Overall, **Hybrid** provides the best overall fit, with **KRR** performing robustly on distributional moments.

7 Conclusion

This project explored the use of *path signatures* as a representation tool for generative modeling of financial time series. Two signature-based models—**BootstrapPathwise** and **KRRSignature**—were provided by the sponsor, while we developed a new hybrid variant, the **HybridKR-RBootstrap** model, combining nonparametric residual sampling with kernel ridge regression on signatures for drift estimation. For benchmarking, we included classical econometric models, namely **ARIMA** and **GARCH**.

Using synthetic datasets generated from four distinct stochastic dynamics (GBM, CEV, Merton jump–diffusion, and Variance–Gamma), we evaluated all models through (i) signature-based Maximum Mean Discrepancy (MMD), (ii) Kolmogorov–Smirnov tests on per-path moments (mean, variance, skewness, kurtosis), and (iii) qualitative comparisons of empirical distributions (histograms and ECDFs). Results show that while the hybrid model often improves stability and realism compared to pure bootstrap generation, performance varies across dynamics: the KRR-based approaches perform better for diffusion-type processes (GBM, CEV), whereas bootstrap methods are more flexible under jump or heavy-tailed distributions.

Overall, the experiment demonstrates that signature representations provide a promising, model-agnostic framework for generative modeling of financial time series. They capture path-wise temporal structure in a way that complements classical parametric methods. Future work could extend these models to multivariate settings, introduce rough-path-inspired stochastic volatility components, and evaluate real market data calibration and conditional path generation.

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A Additional Qualitative Figures

A.1 GBM

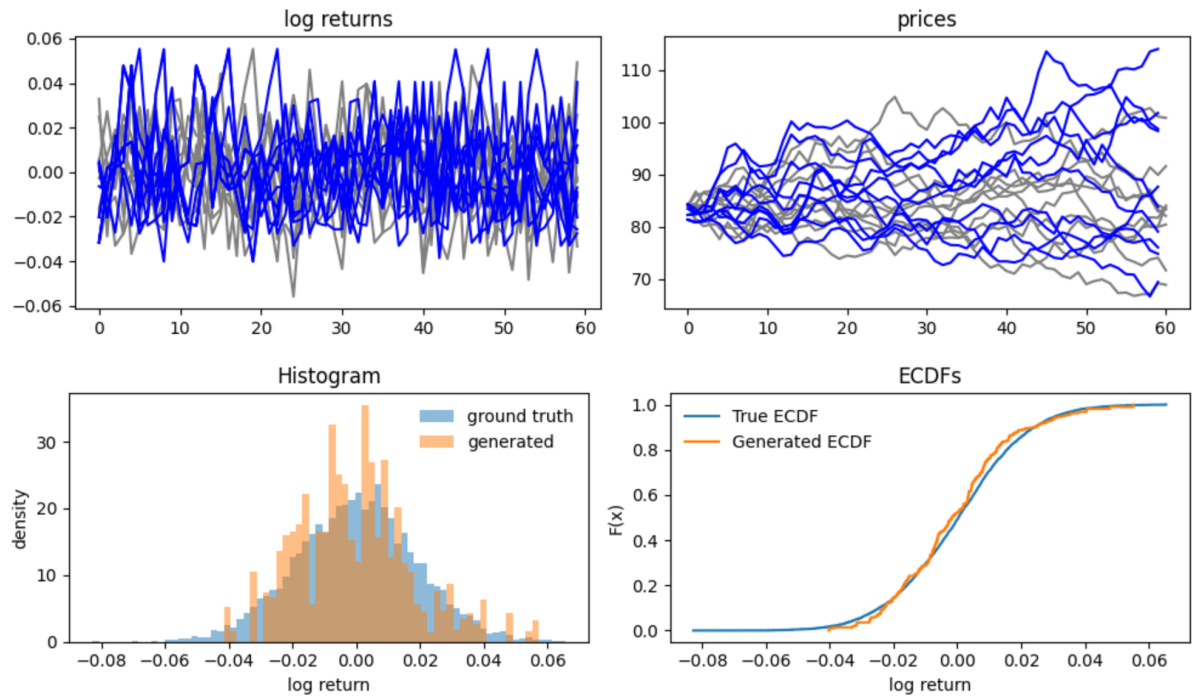


Figure 2: GBM – Bootstrap

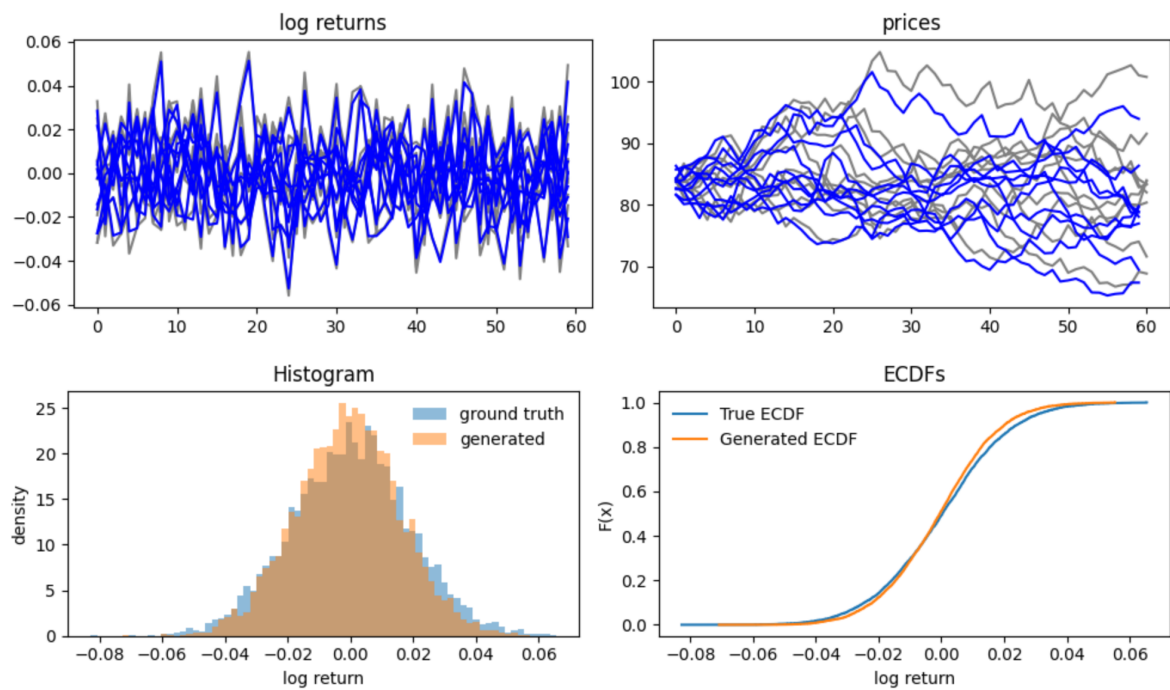


Figure 3: GBM – KRR

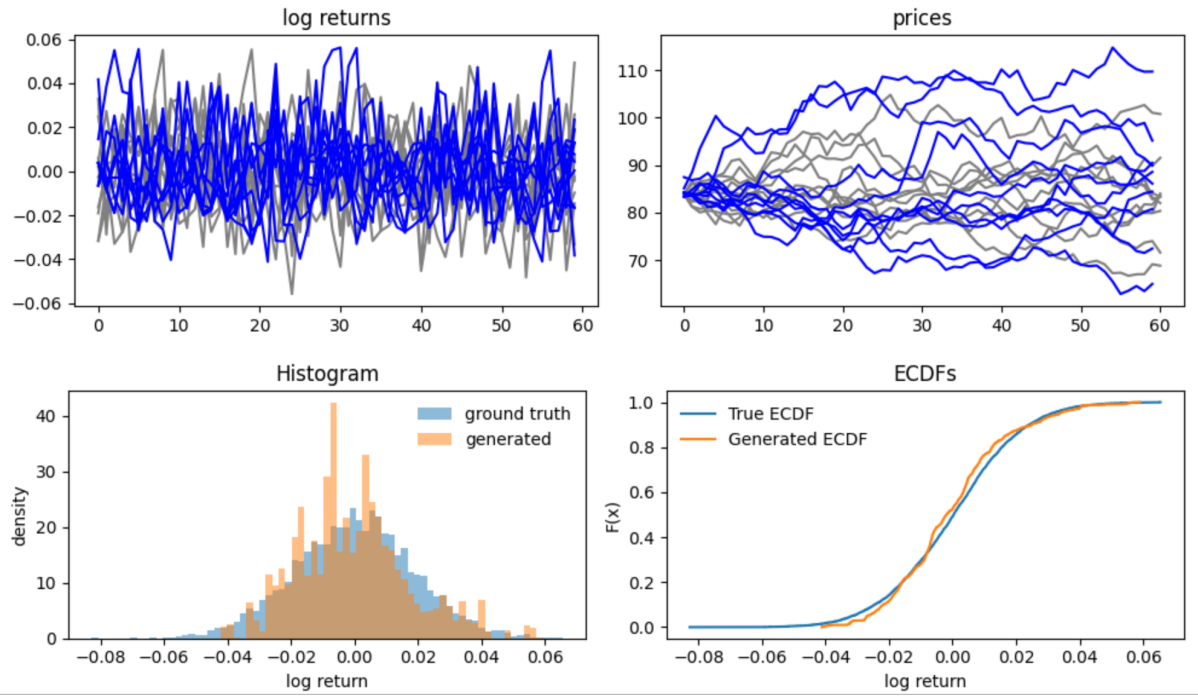


Figure 4: GBM – Hybrid KRR

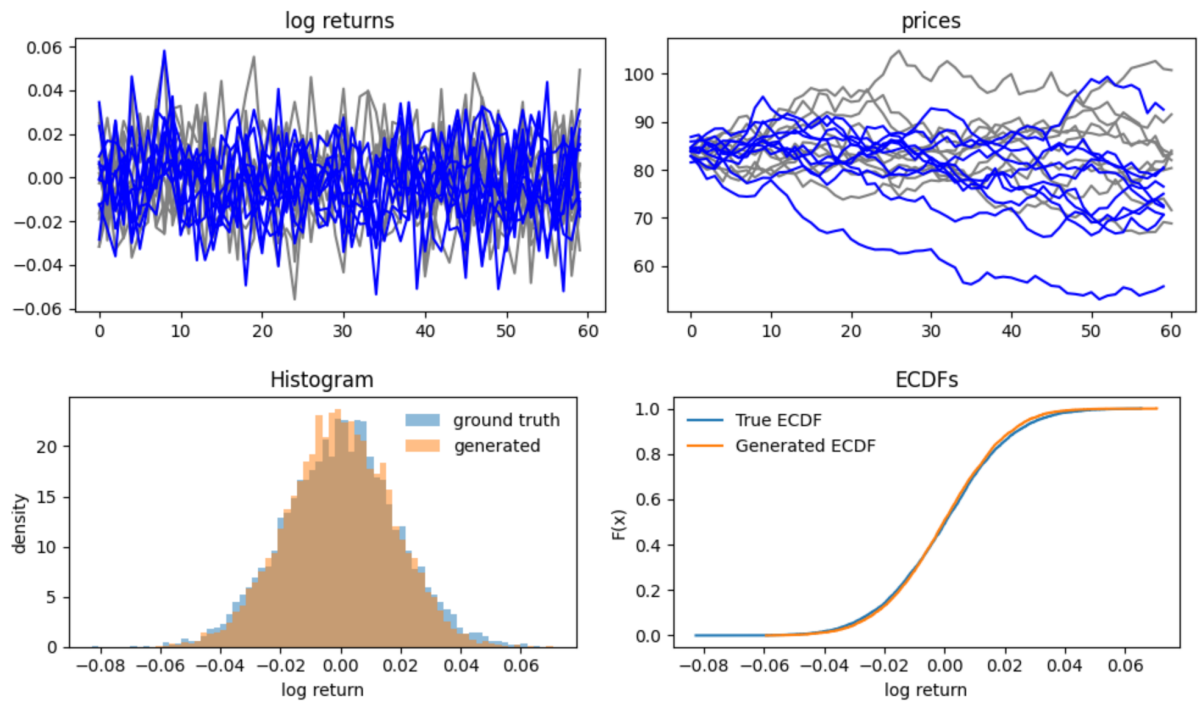


Figure 5: GBM – ARIMA

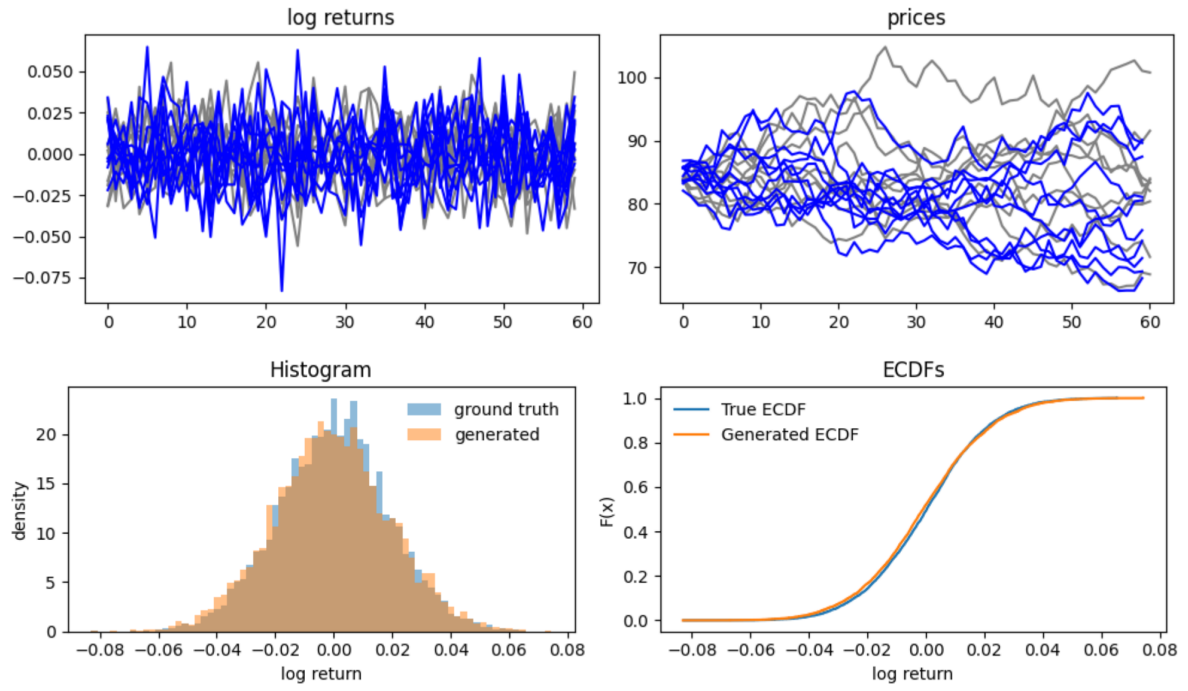


Figure 6: GBM – GARCH

A.2 CEV

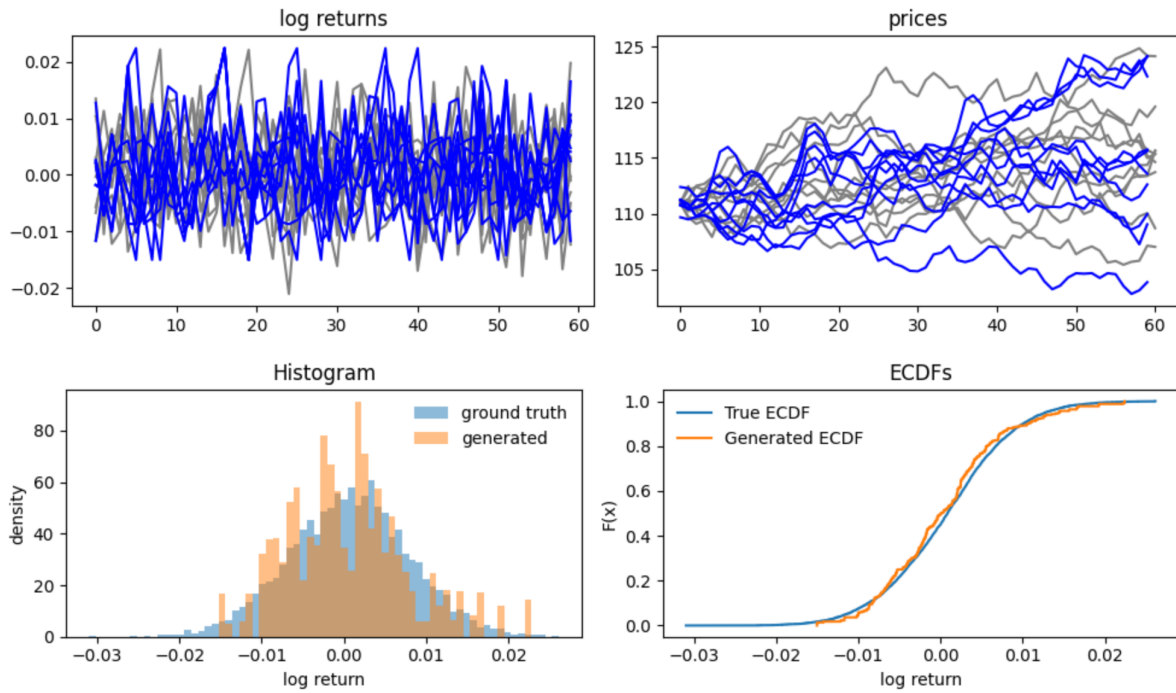


Figure 7: CEV – Bootstrap

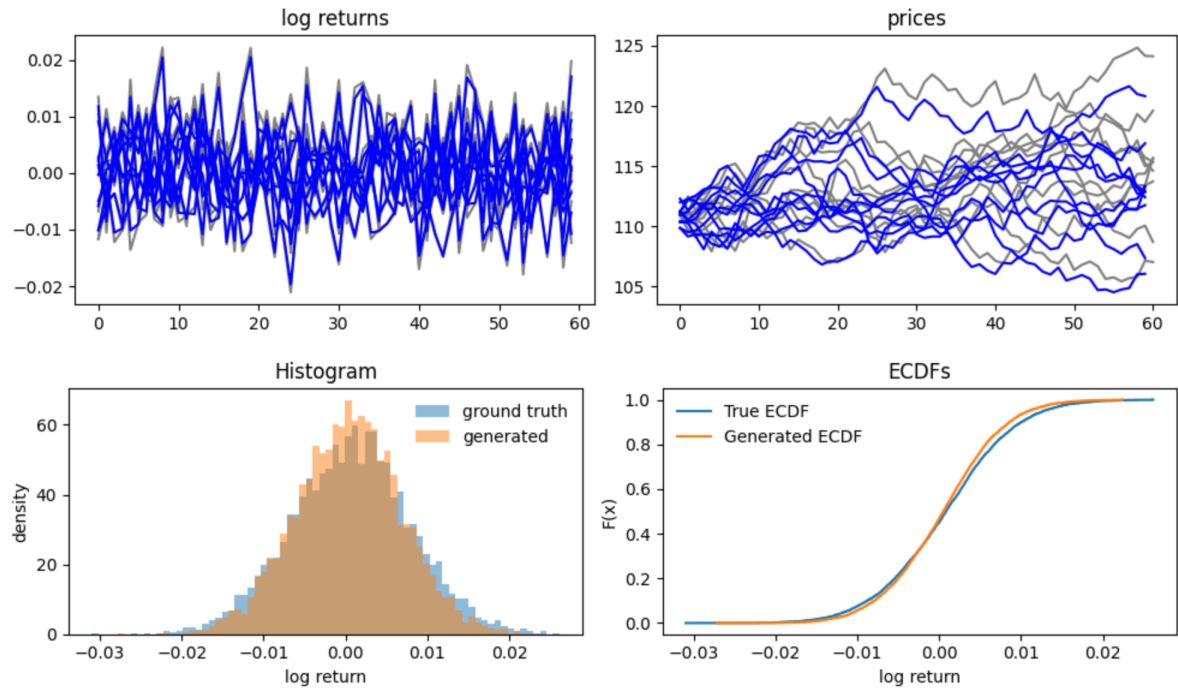


Figure 8: CEV – KRR

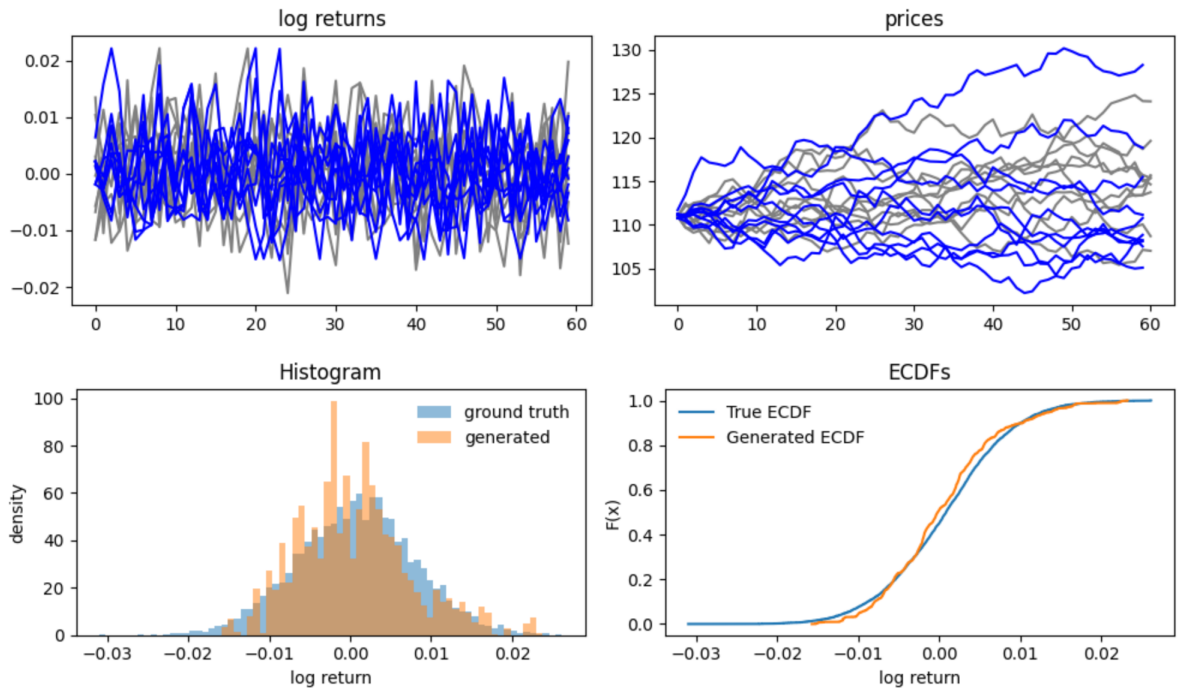


Figure 9: CEV – Hybrid KRR

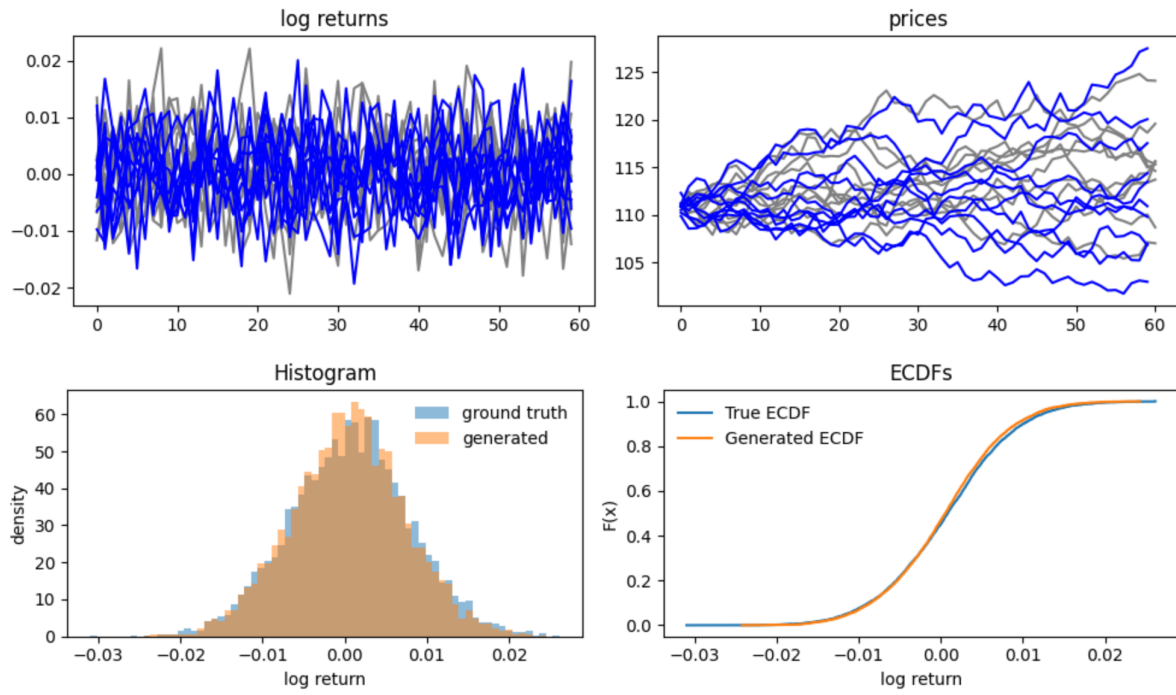


Figure 10: CEV – ARIMA

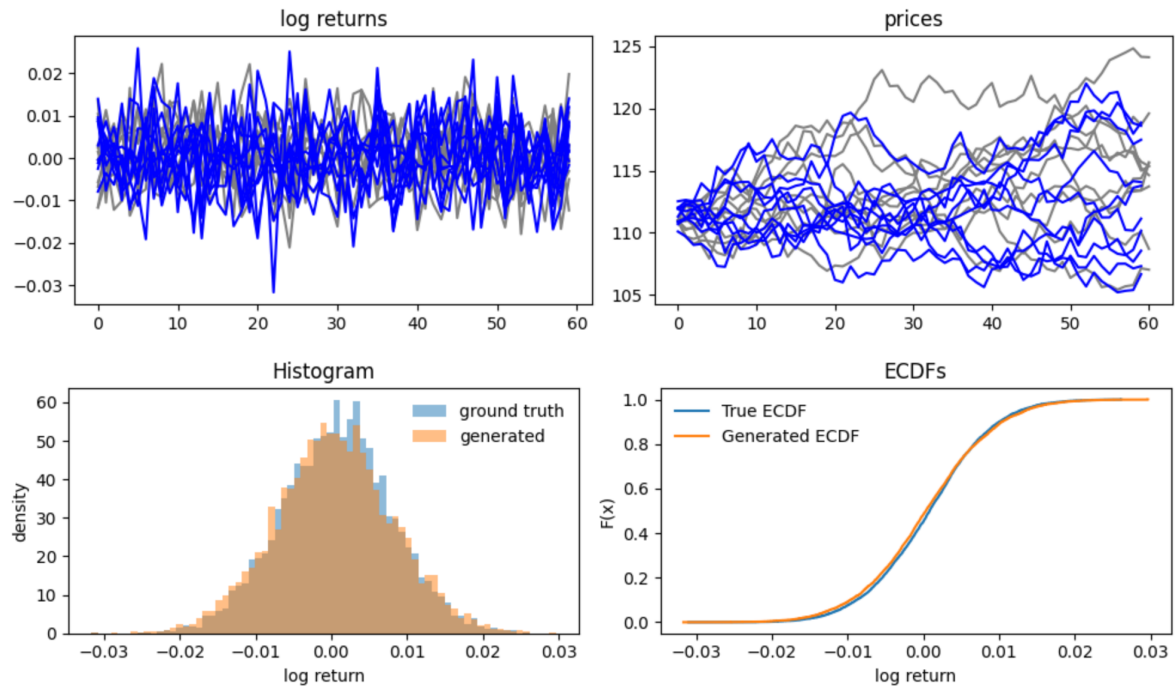


Figure 11: CEV – GARCH

A.3 Merton

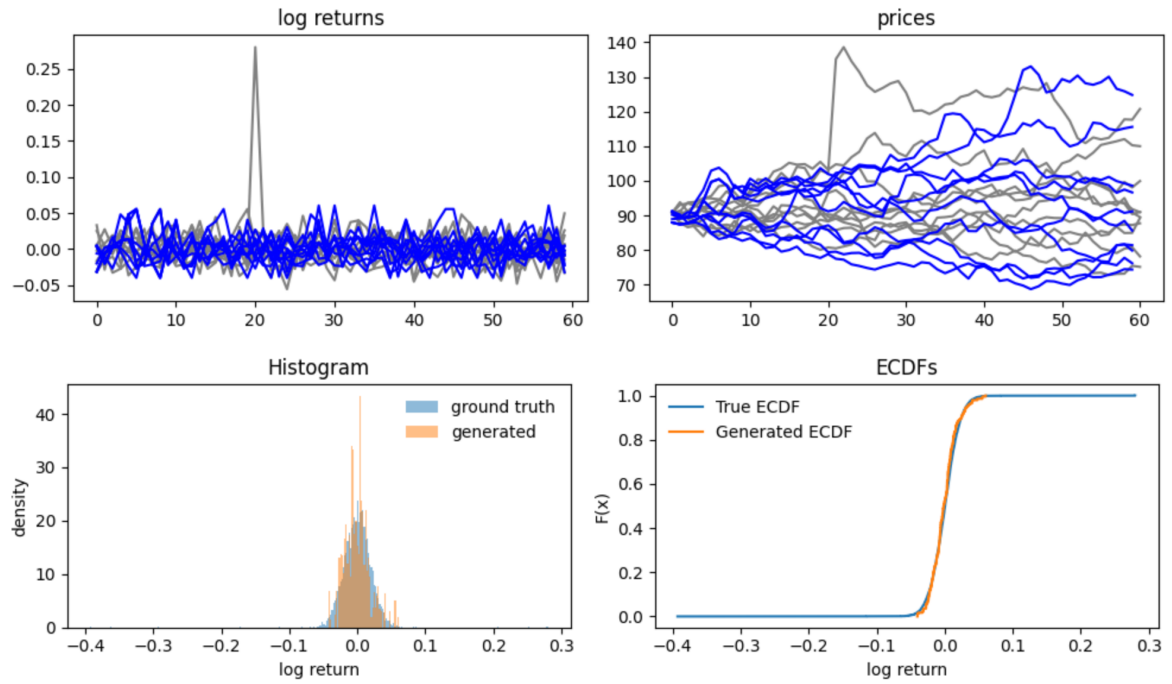


Figure 12: Merton – Bootstrap

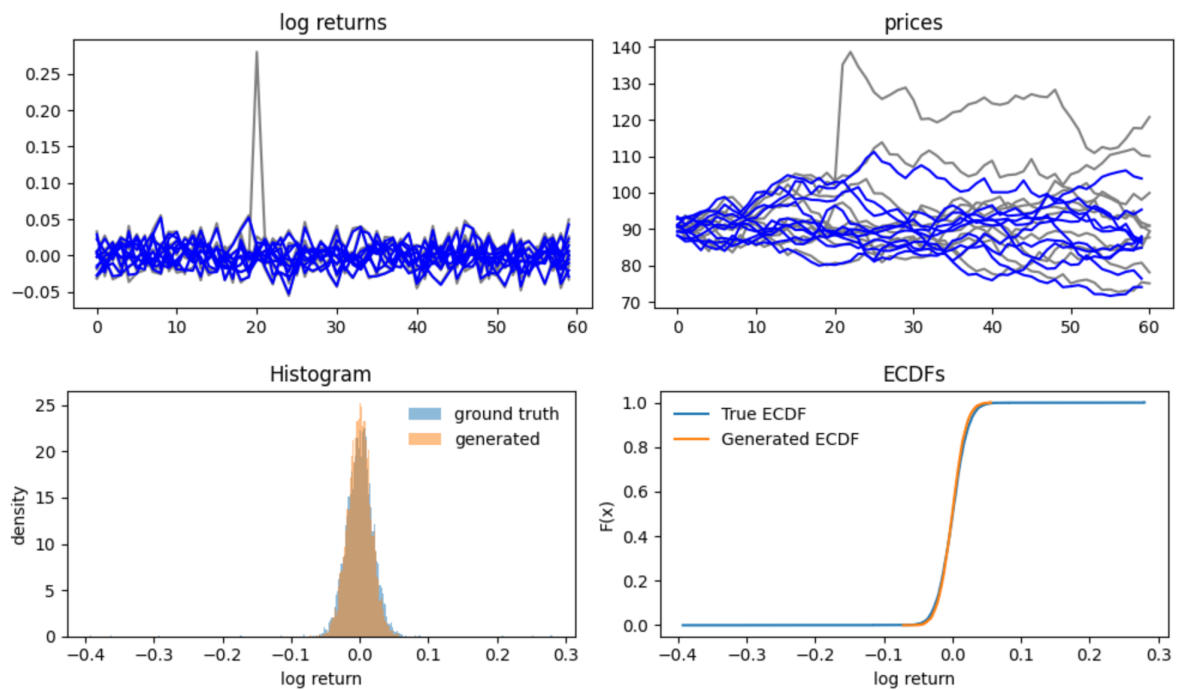


Figure 13: Merton – KRR

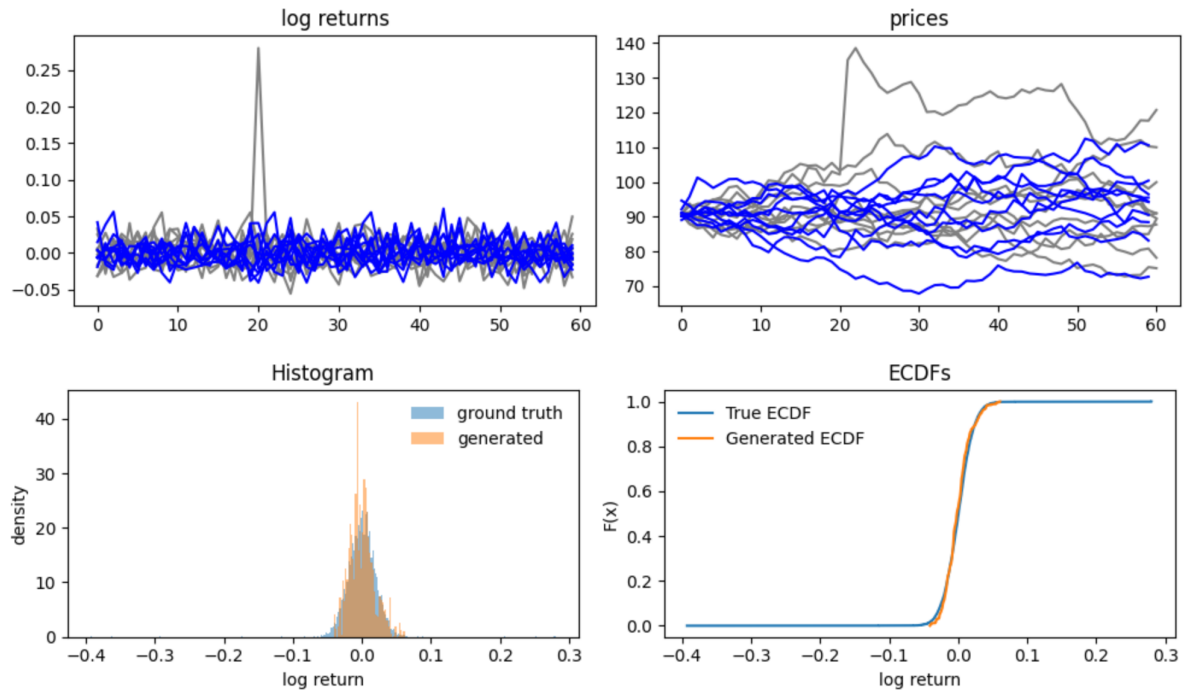


Figure 14: Merton – Hybrid KRR

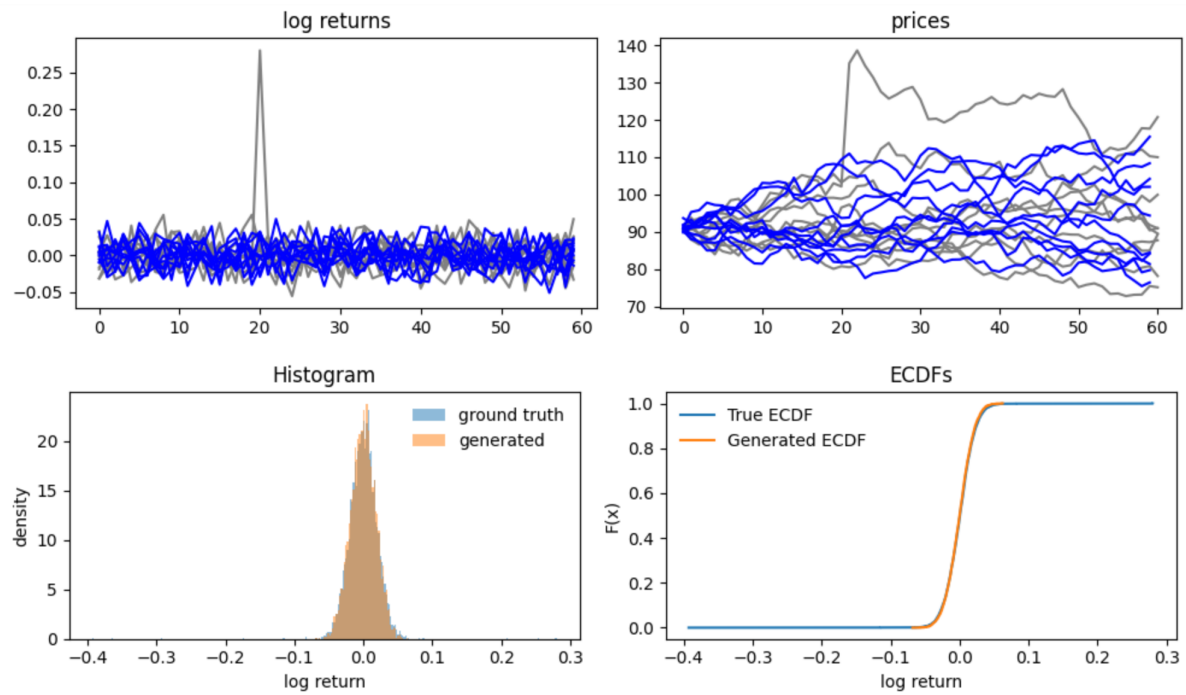


Figure 15: Merton – ARIMA

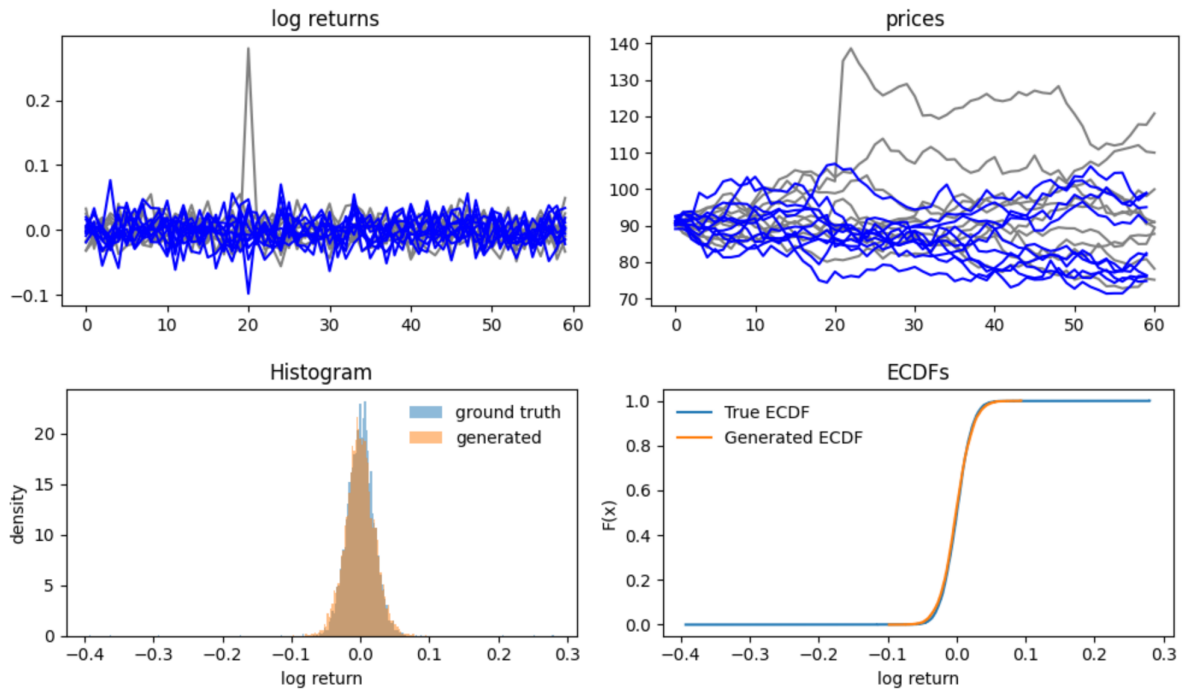


Figure 16: Merton – GARCH

A.4 Variance–Gamma

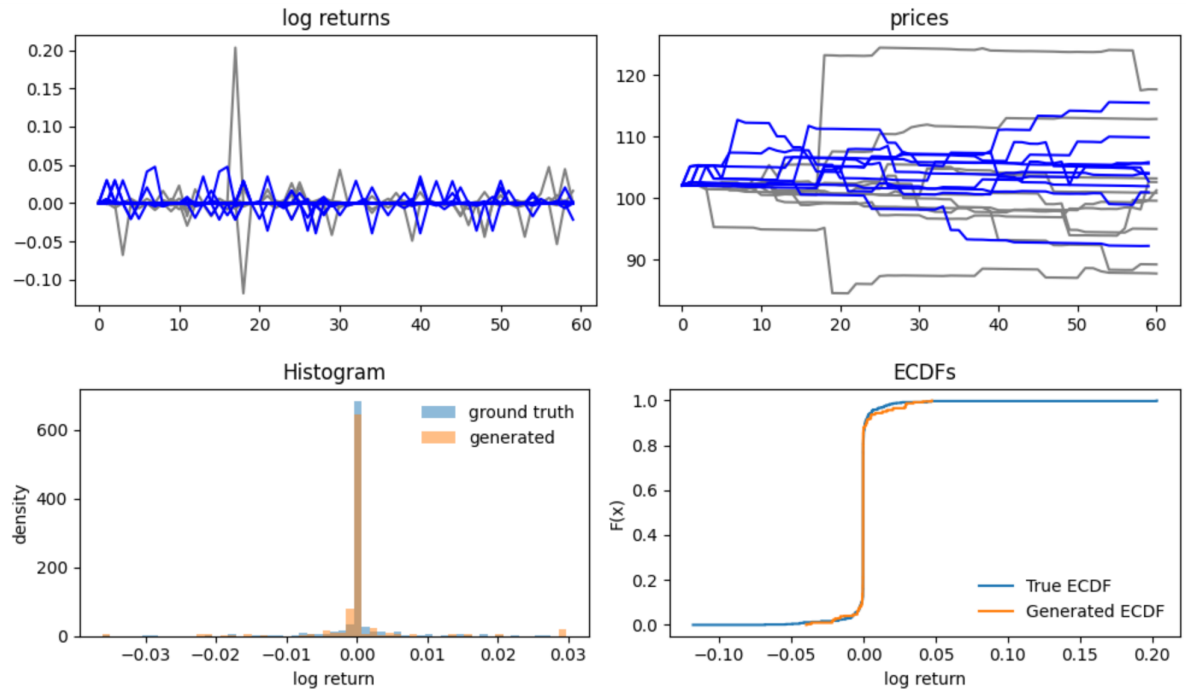


Figure 17: VG – Bootstrap

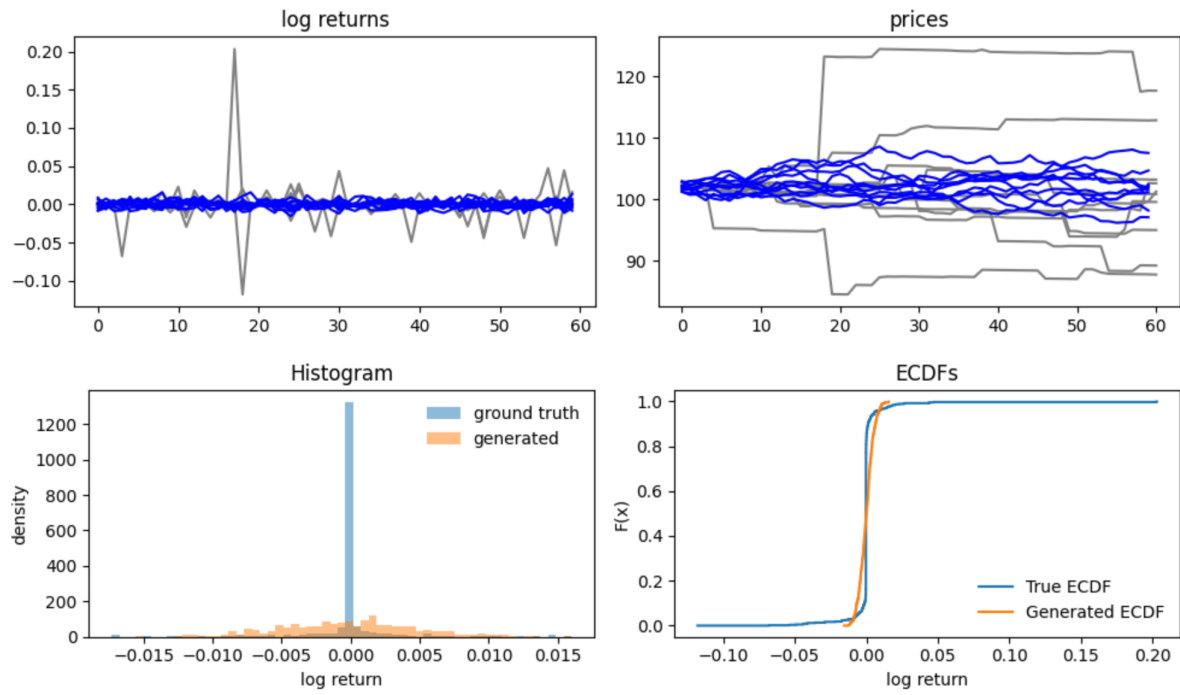


Figure 18: VG – KRR

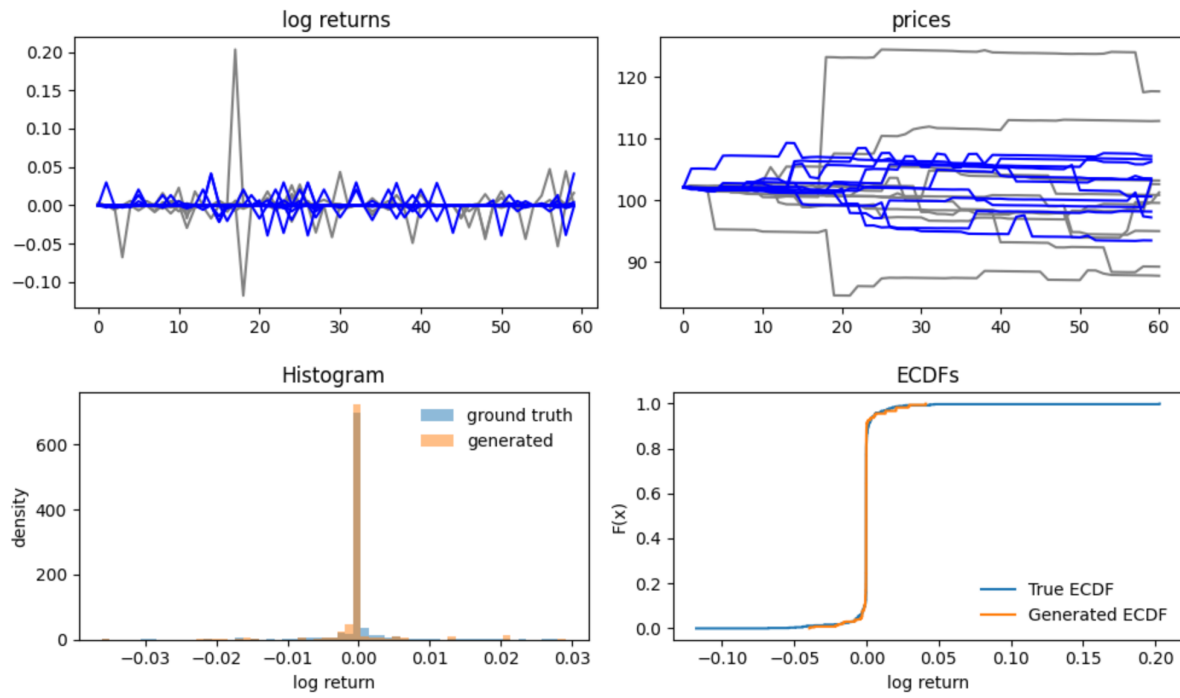


Figure 19: VG – Hybrid KRR

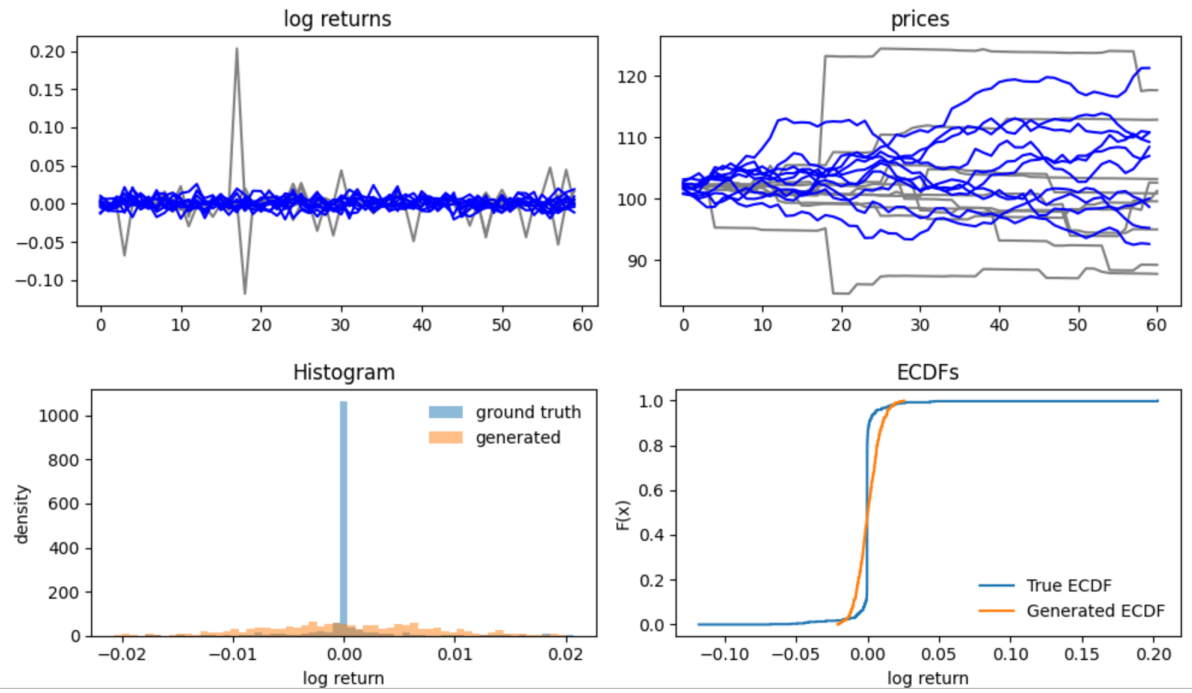


Figure 20: VG – ARIMA

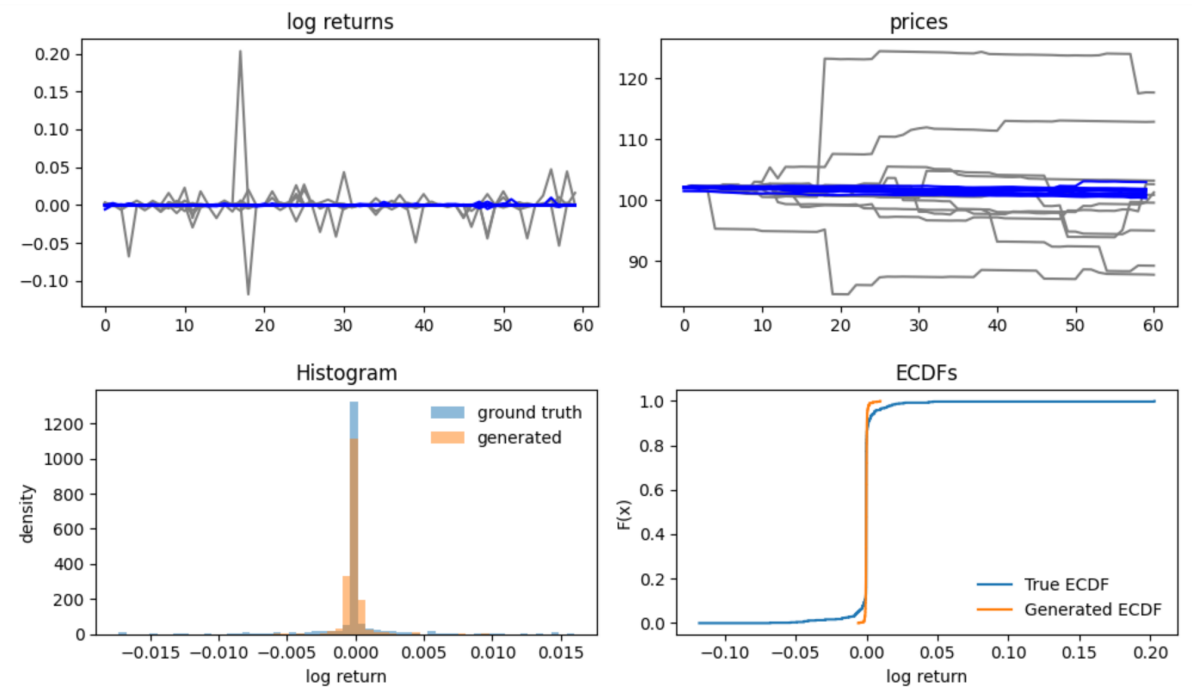


Figure 21: VG – GARCH

A.5 Heston

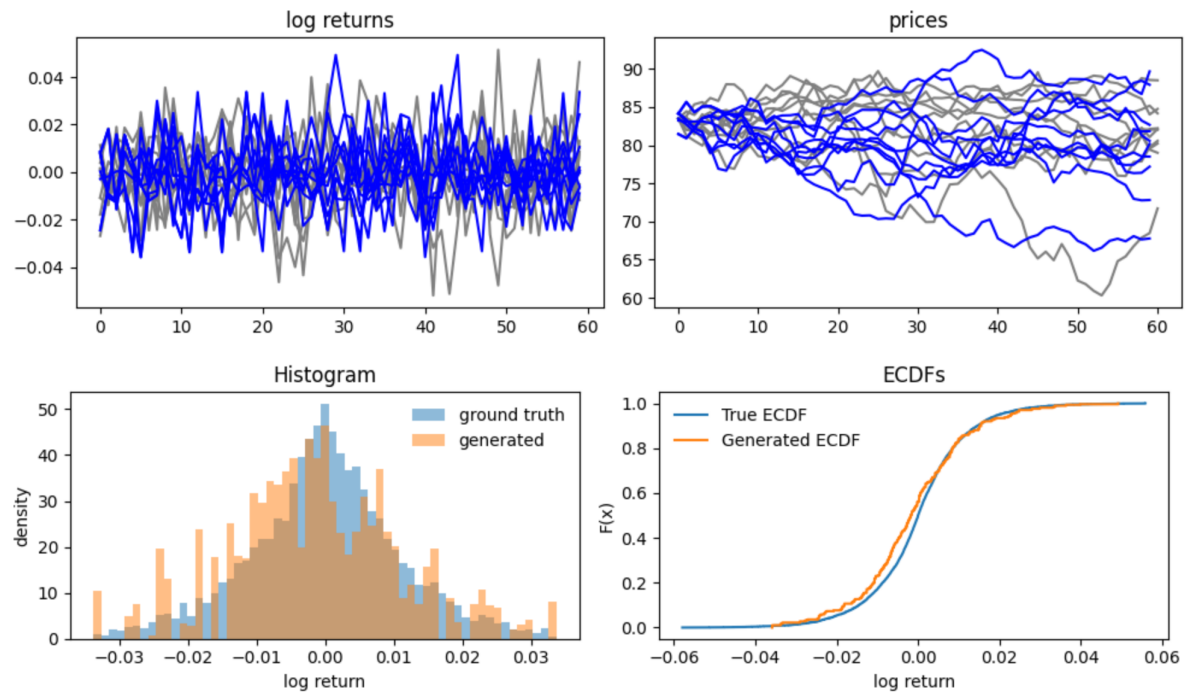


Figure 22: Heston – Bootstrap

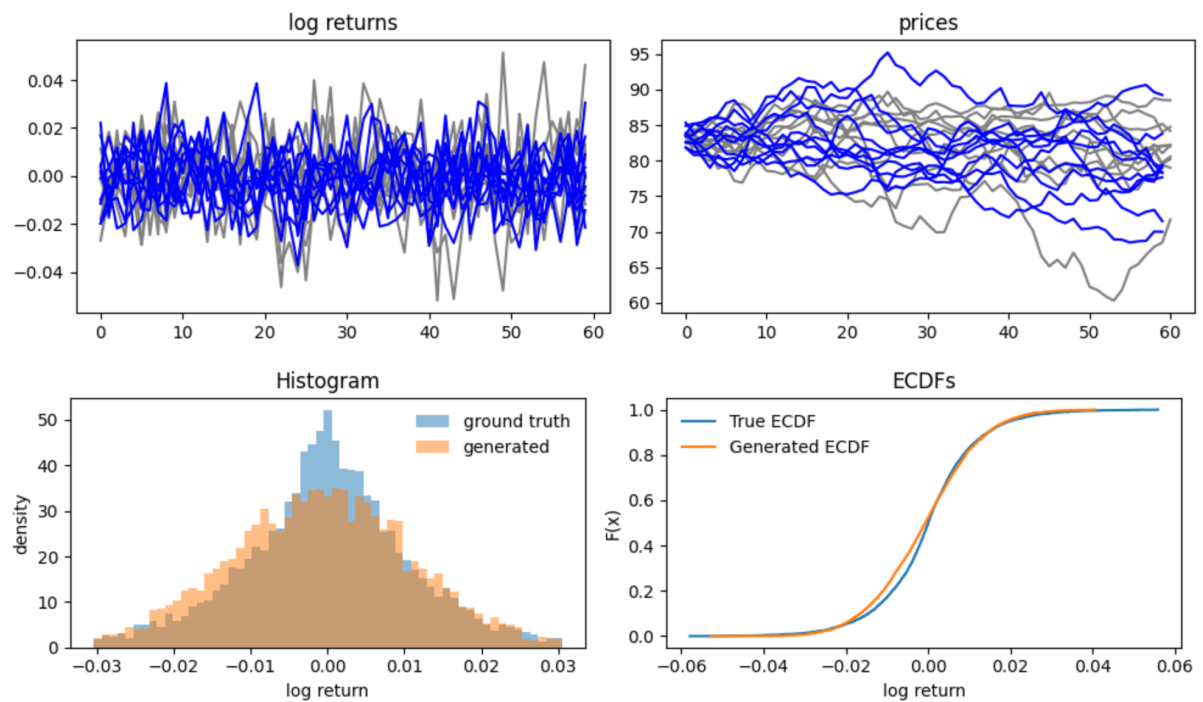


Figure 23: Heston – KRR

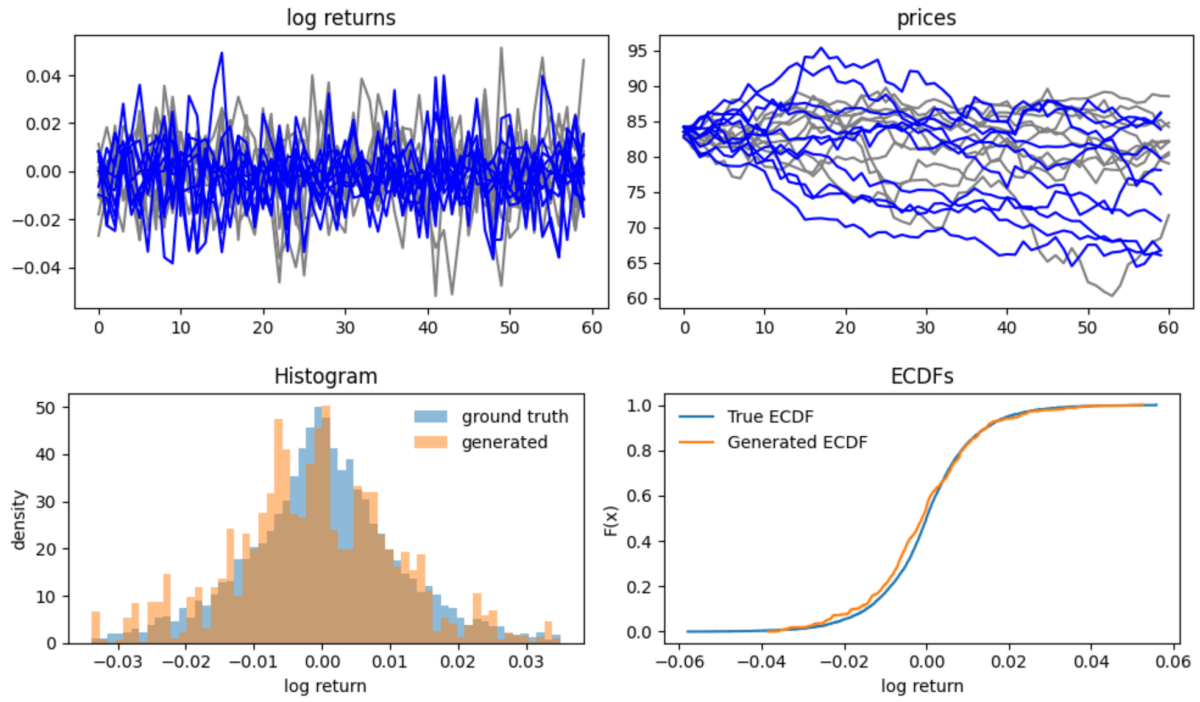


Figure 24: Heston – Hybrid KRR

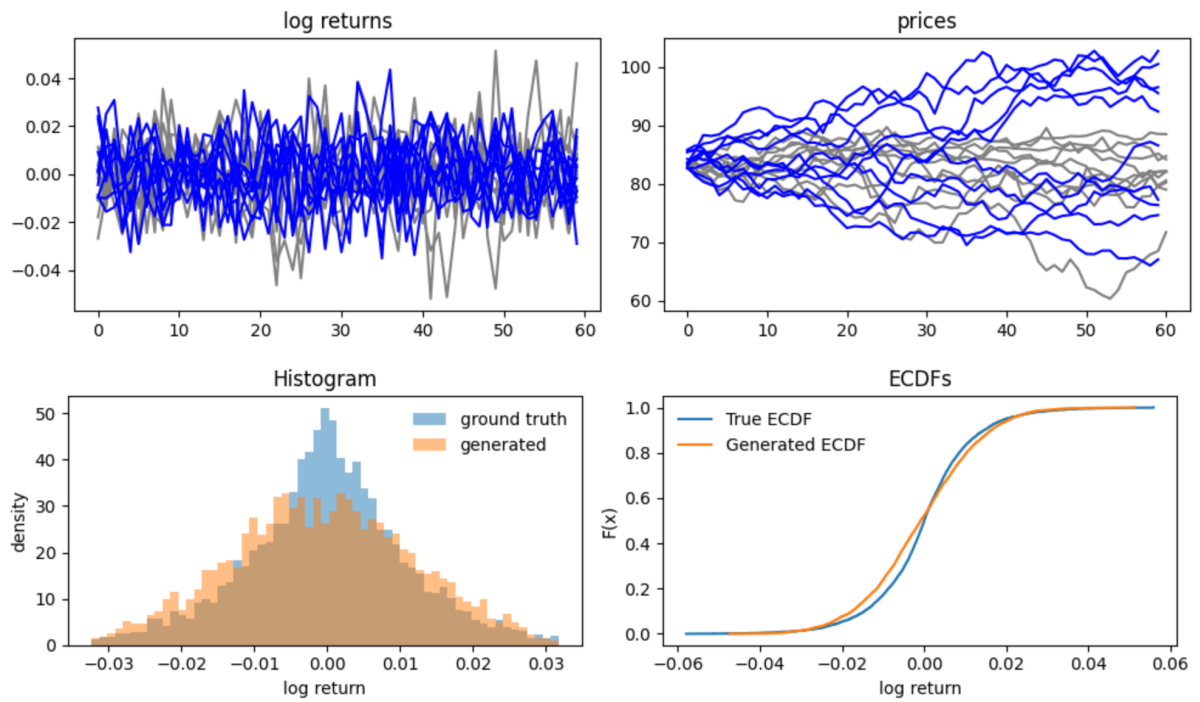


Figure 25: Heston – ARIMA

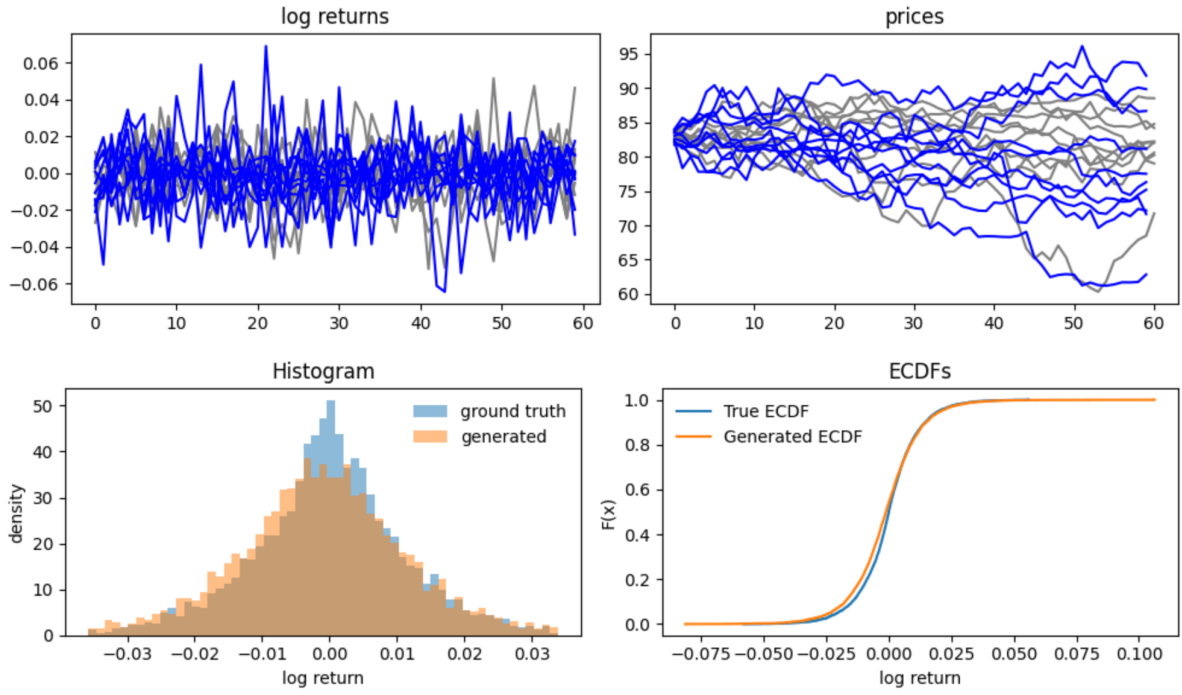


Figure 26: Heston – GARCH

A.6 Garch-return

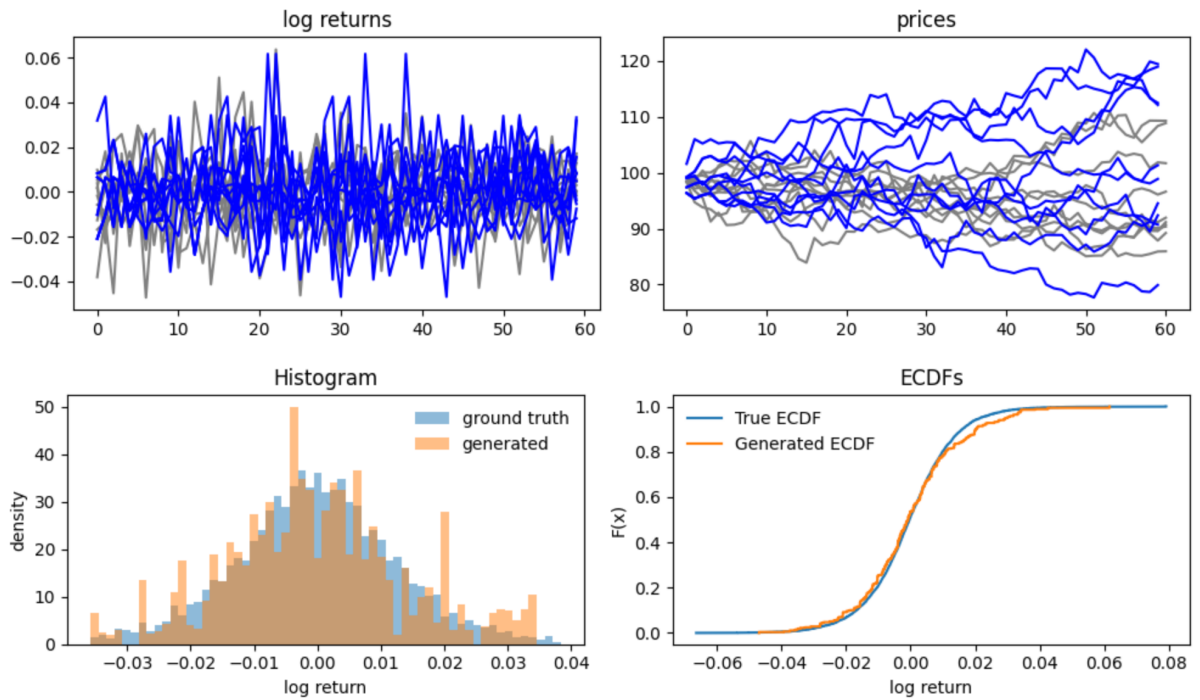


Figure 27: Garch – Bootstrap

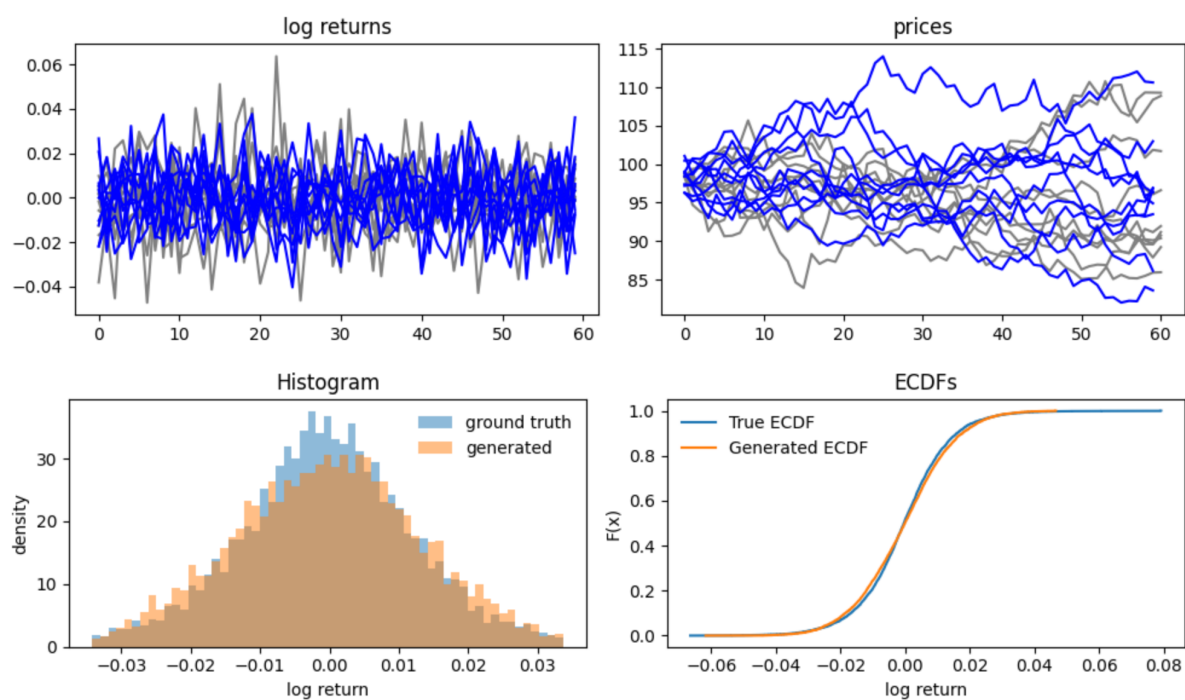


Figure 28: Garch – KRR

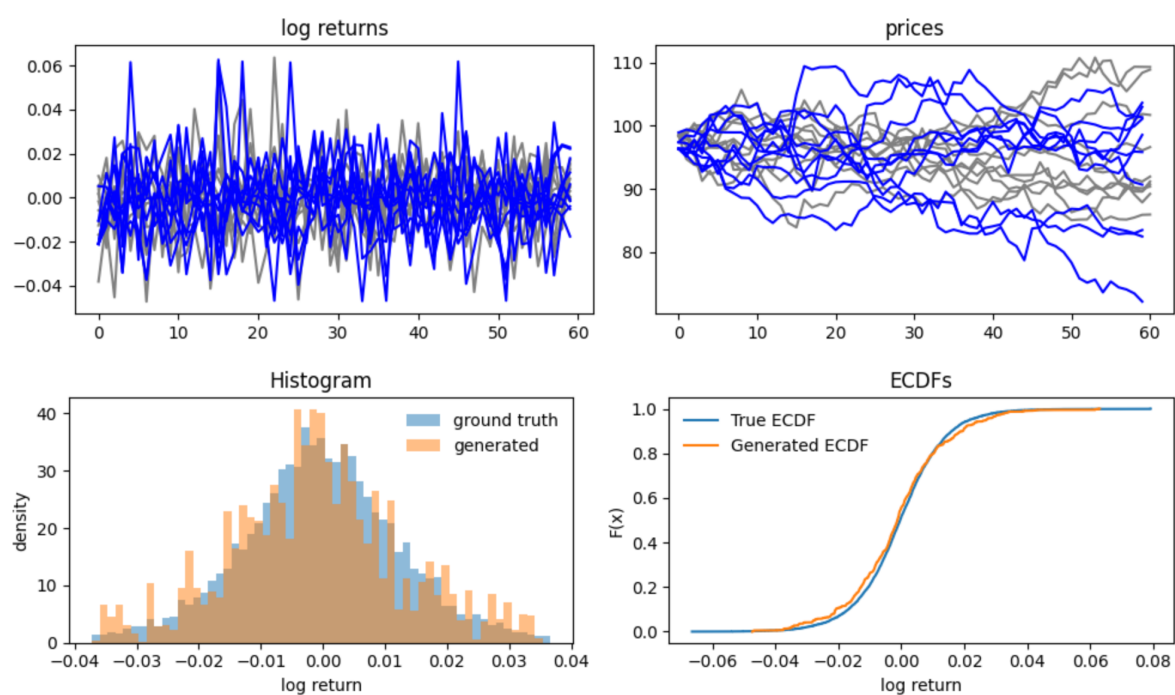


Figure 29: Garch – Hybrid KRR

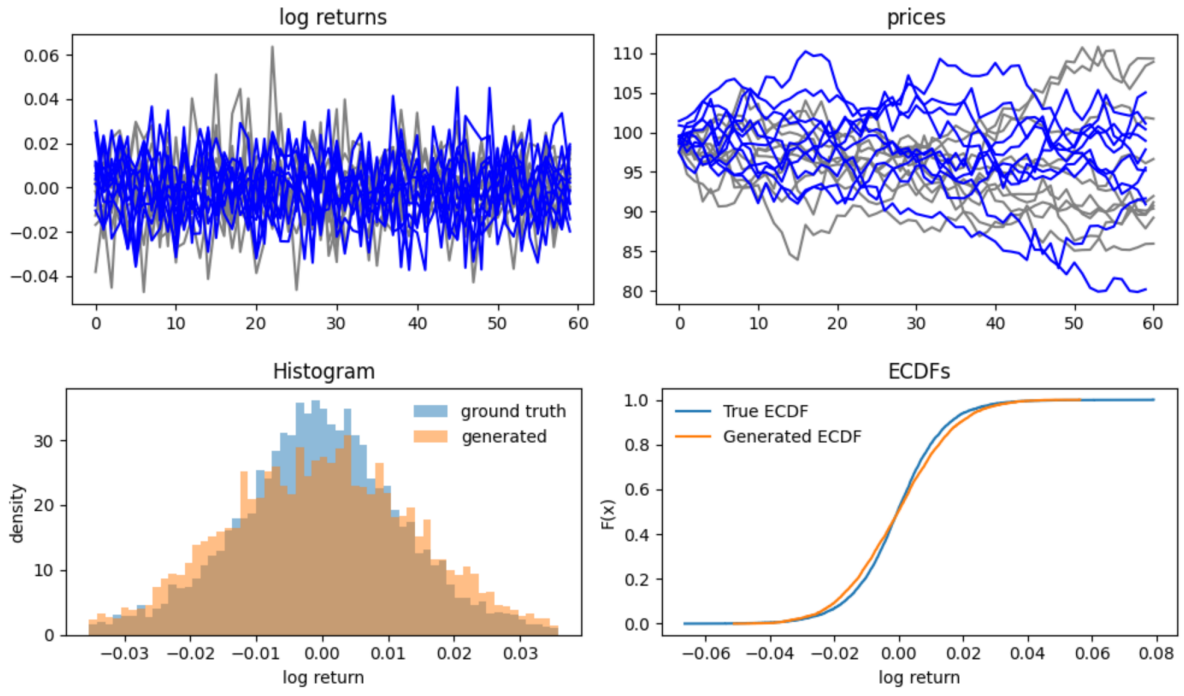


Figure 30: Garch – ARIMA

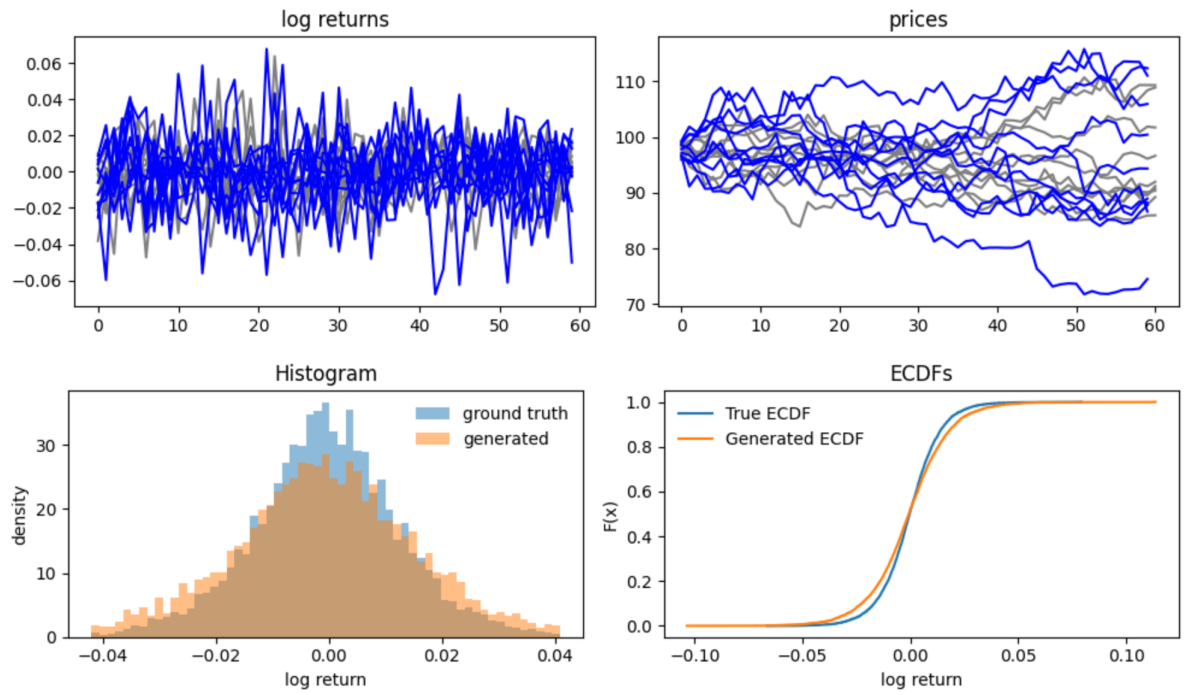


Figure 31: Garch – GARCH